Dynamic Nonlinear Seismic Response Analysis of the RC Upright Construction Method

Report

June 22, 2005

A Prominent Research Institution

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1. Overview

The purpose of this analysis is to analytically verify the structural integrity of a two-story building utilizing the RC Upright Construction Method (reinforced concrete structure) when subjected to major earthquakes corresponding approximately to Seismic Intensity Levels 6 to 7.

The RC Upright Construction Method is an entirely new construction system consisting of a reinforced concrete structure with a mat foundation (450 mm or greater in thickness) and one-directional exterior and partition walls (300 mm or greater in thickness), combined with timber floors, walls, and roof structures constructed using the 2×4 method. The RC Upright Construction Method possesses outstanding characteristics in adaptability, fire resistance, seismic performance, condominium ownership compatibility, sound insulation, renovation flexibility, and architectural design freedom. A patent application for this construction system is currently being filed by your company.

In this analysis, the two-story RC Upright Construction Method building is modeled using three-dimensional thick-shell elements incorporating reinforcing bars, and a dynamic nonlinear seismic response analysis is performed to evaluate its structural integrity under earthquake loading.

The general-purpose finite element analysis program ABAQUS Ver. 6.5 is used in this analysis.

2. Analysis Conditions

2.1 Subject of Analysis

Figure 2.1 shows an example of a building constructed using the RC Upright Construction Method (Takashima Apartment Building). Figure 2.2 presents an overview of the two-story RC Upright Construction Method building. Figure 2.3 shows the drawings of the subject building. As shown in Figure 2.3, the subject of this analysis consists of a single reinforced concrete structural system comprising a mat foundation, two exterior walls, and two partition walls.

The principal structural dimensions of the subject of analysis are as follows: wall height 5,400 mm, thickness 300 mm, width 10,310 mm; foundation slab 13,500 mm × 10,310 mm, thickness 500 mm.

The reinforcing bars are SD295 grade steel with lap-splice connections. The arrangement of the wall main reinforcement (vertical reinforcement) is D16 @160W for the portion above GL+2150 mm, and D13/D16 @80W for the portion below GL+2150 mm; the horizontal reinforcement is D10 @100W. The effective depth of the wall is d = 240 mm. The main reinforcement of the foundation slab is D13/D16 @160W, the stirrups are D13 @190W, and the effective depth of the foundation slab is d = 410 mm. The unit system is the SI system (N, mm, sec).

2.2 Analysis Code

The general-purpose finite element analysis program ABAQUS is used in this analysis. ABAQUS consists of ABAQUS/Standard (general-purpose FEA program), ABAQUS/Explicit (explicit dynamic FEA program), and ABAQUS/CAE (complete ABAQUS operating environment including pre/post-processing tools). ABAQUS/Standard is developed for advanced engineering analyses and practical applications. At present, ABAQUS is the most widely established analysis code both in Japan and internationally, applied in automotive, mechanical, aerospace, nuclear, heavy industry, building/civil engineering, and chemical engineering. It is used for verification of structural safety and integrity in structural design, and is regarded as a highly reliable code. ABAQUS features an extensive element library, a large collection of verification examples, and advanced automatic increment control capabilities.

2.3 Analytical Model

(1) Reinforced Concrete Foundation and Walls

As an initial stage of this study, the two-story RC Upright Construction Method building shown in Figure 2.3 is modeled using three-dimensional thick-shell elements. The concrete portions of the walls and foundation slab are modeled at the mid-surface of the wall and slab thicknesses using three-dimensional thick-shell elements. The reinforcing bars are modeled using the ABAQUS reinforcement element function (*REBARLAYER data). By combining these models, the walls and foundation slab are analyzed as a reinforced concrete structure.

In the three-dimensional shell model, only the main reinforcing bars (vertical reinforcement) and horizontal reinforcing bars can be modeled. However, the hoop reinforcement in the thickness direction, which connects the horizontal reinforcing bars, cannot be modeled. Since the wall height (~5,400 mm) is large relative to the wall thickness (300 mm), the expected failure mode during an earthquake is considered to be flexural failure, and the contribution of hoop reinforcement against wall buckling and shear failure is considered relatively small.

In addition, wall segments around openings such as windows and doors are conservatively neglected, and the openings are modeled as full-height vertical slits extending over the entire wall height.

(2) Timber Floors, Walls, and Roof Structure

The timber floor, wall, and roof components are not explicitly modeled in the present analysis; their masses are incorporated as additional loads. The additional loads for the timber floors, walls, and roof structures are described in Section 2.6.

(3) Soil

The soil is modeled using horizontal and rotational soil springs. The soil spring constants are described in Section 2.7.

2.4 Material Properties

(1) Concrete

Since ABAQUS/Standard Version 6.3, the Concrete Damaged Plasticity Model has been introduced to improve the accuracy of structural analyses of reinforced concrete structures. In this analysis, the Concrete Damaged Plasticity Model is adopted as the constitutive model for concrete. A detailed description of this model is provided in Appendix 4.

The concrete used for the walls and foundation of the two-story RC Upright Construction Method building is normal-weight concrete with a specified design compressive strength of Fc = 21 (N/mm^2). In this analysis, the concrete of the walls and foundation slab is modeled as a nonlinear material that accounts for concrete cracking. The material properties used for the concrete are as follows. Figure 2.26 shows the uniaxial stress–strain relationship for concrete. As shown in the figure, the compression side is modeled as a bilinear relationship, and the tension side as a trilinear relationship.

- Young's Modulus: **EC = 21,700 N/mm^2**
- Poisson's Ratio: **ν = 0.167**
- Compressive Strength: **σU = 21 × 0.85 / 1.3 = 13.731 N/mm^2**
- Ultimate Compressive Strain: **εCU' = 0.0035**
- Tensile Strength: **σT = 3.882 N/mm^2**

(**ftk = 0.23fck^**(2/3) **+ Bond Strength 0.28fck^**(2/3))

The tension stiffening behavior governing the post-cracking response is shown in Figure 2.26(b). Figure 2.27 illustrates the cyclic characteristics of the Concrete Damaged Plasticity Model. For the ABAQUS Concrete Damaged Plasticity Model, the following parameters are adopted: Dilation Angle φ = 35°, Viscosity Parameter $\mu = 1.0 \times 10^{-5}$, Compression Stiffness Recovery Factor WC = 1.0 (default), Tension Stiffness Recovery Factor WT = 0.0 (default). The concrete material properties are defined using the following ABAQUS options:

```
*CONCRETE DAMAGED PLASTICITY
*CONCRETE COMPRESSION HARDENING
*CONCRETE TENSION STIFFENING, TYPE=STRAIN
*CONCRETE TENSION DAMAGE, TYPE=STRAIN
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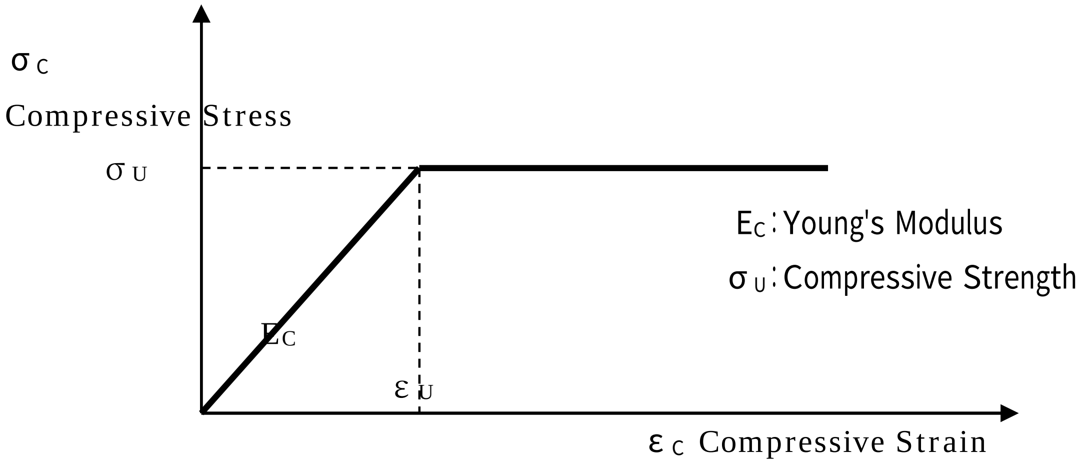
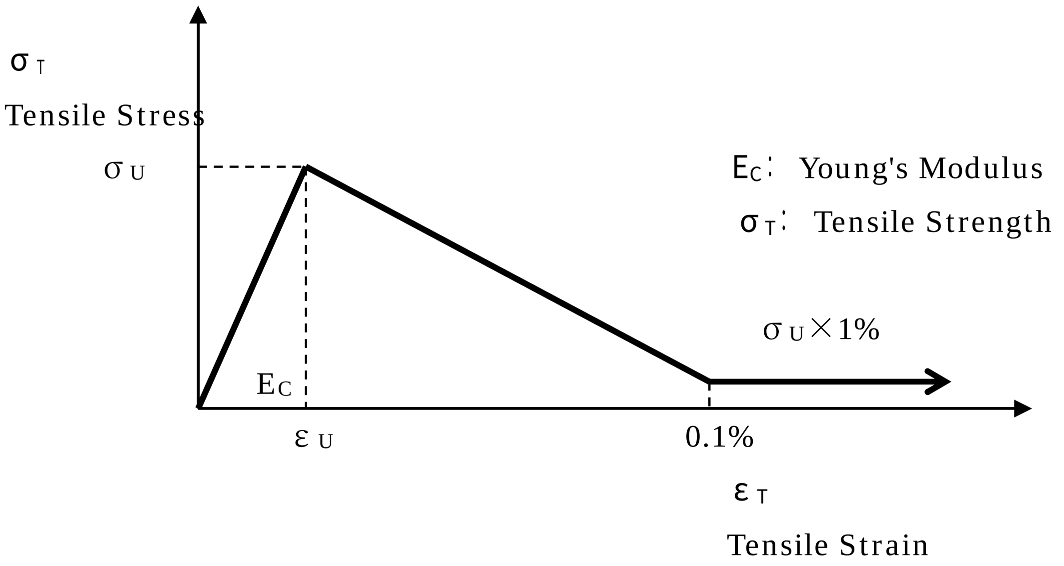
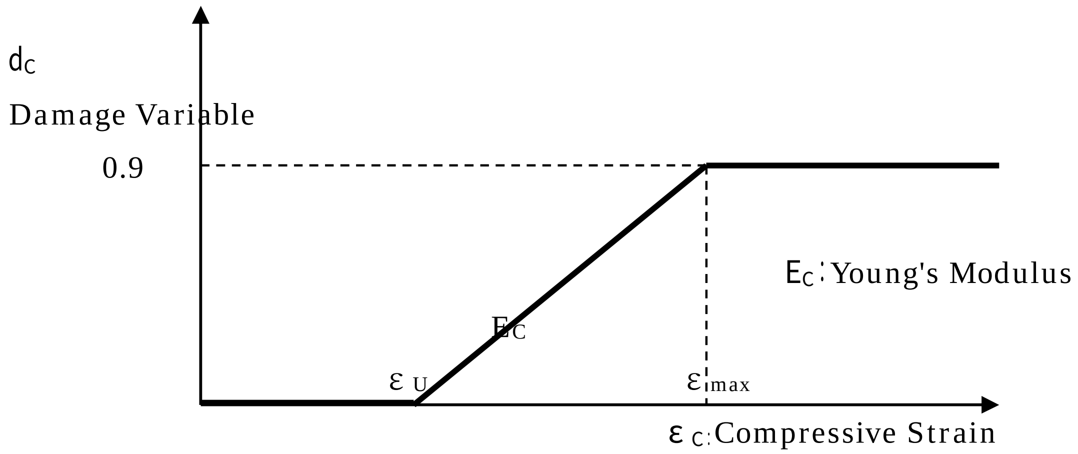


(a) Compression Side (Bilinear Model)

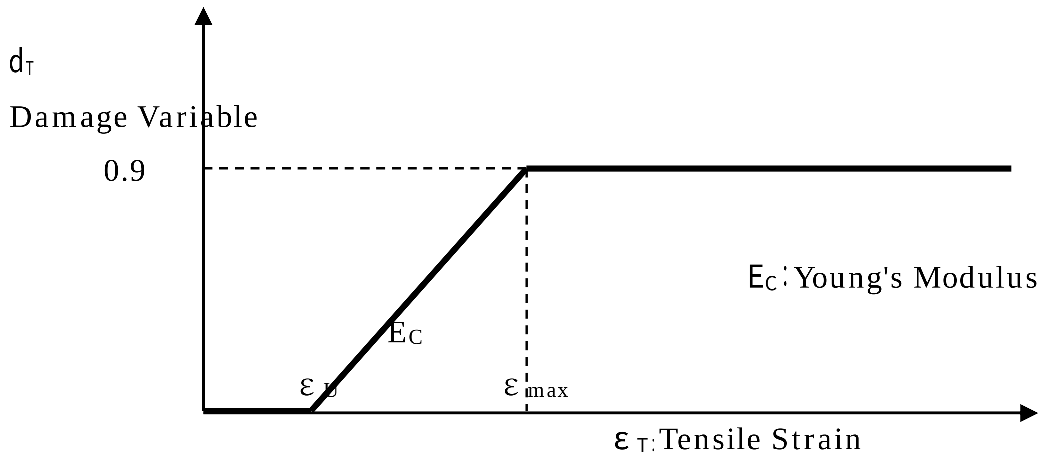


(b) Tension Side (Trilinear Model)

Figure 2.26 Stress–Strain Relationship of Concrete (Uniaxial Loading)



(a) Compression Side



(b) Tension Side

Figure 2.27 Damage Variable–Strain Relationship of Concrete (Uniaxial Loading)

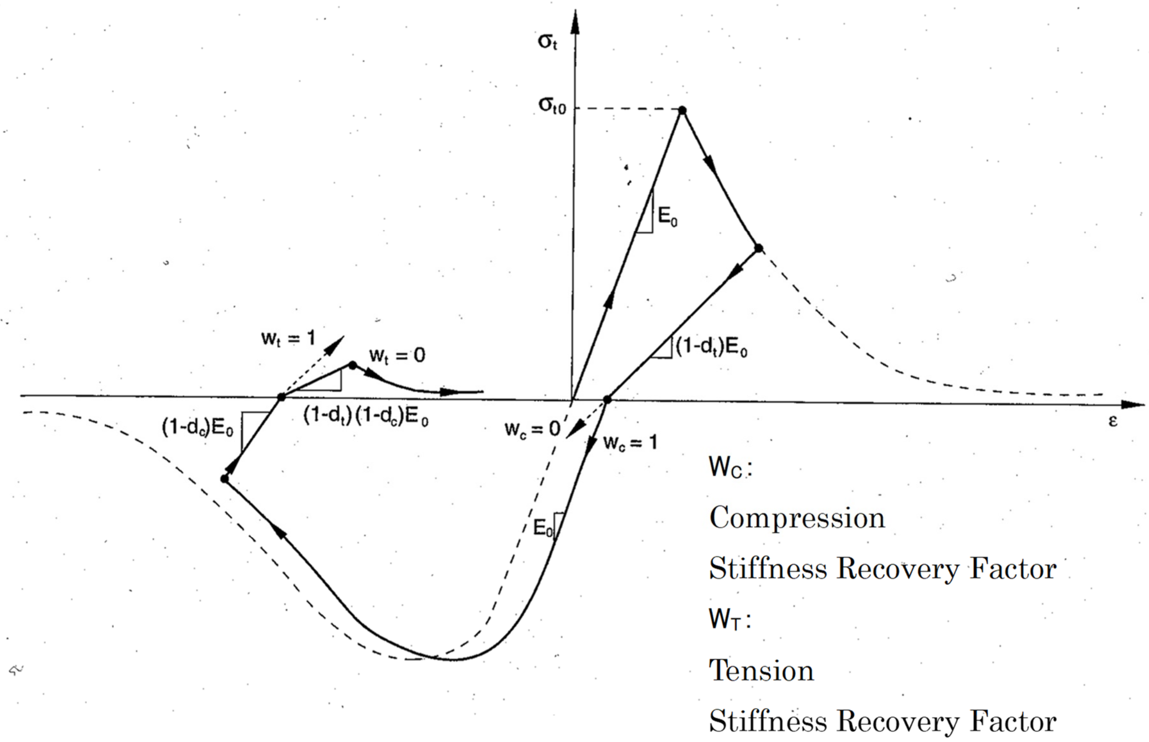


Figure 2.28 Cyclic Behavior of the ABAQUS Concrete Damaged Plasticity Model

(2) Reinforcing Steel

The reinforcing bars are SD295 grade steel with lap-splice connections. The material properties adopted for the reinforcing bars are as follows. Since no prestressing force is applied to the reinforcing bars in the two-story RC Upright Construction Method building, prestressing effects are not considered in this analysis.

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Young's Modulus:  ES = 200,000 (N/mm²)
Poisson's Ratio: ν = 0.3
Yield Stress:    σy = 295 (N/mm²)
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The reinforcing steel is modeled using a bilinear stress–strain relationship, as shown in Figure 2.29.

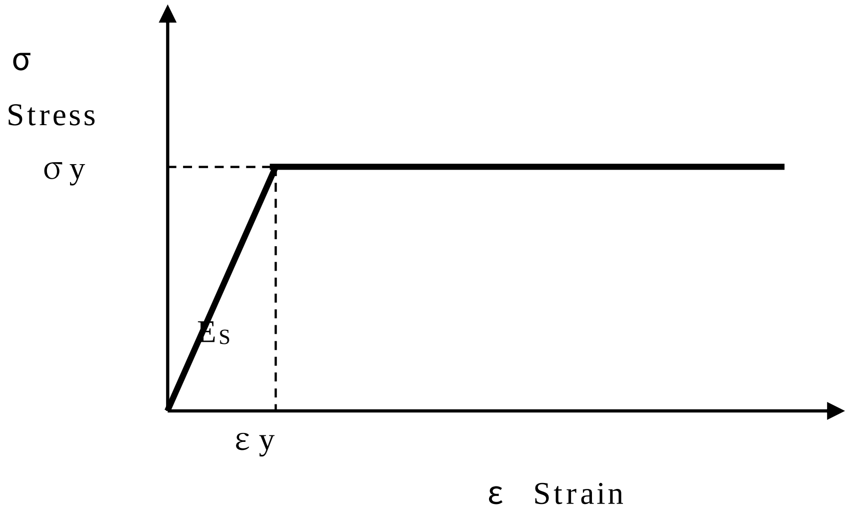


Figure 2.29 Stress–Strain Relationship of Reinforcing Steel

2.5 Boundary Conditions

The RC Upright Construction Method utilizes a mat foundation. The boundary conditions adopted in the seismic response analysis are as follows: under the initial condition (self-weight loading), the vertical displacement of the bottom surface of the mat foundation is supported by vertical springs; during the seismic response analysis, the bottom surface of the mat foundation is supported by horizontal and rotational soil springs. For the natural vibration analysis of the building, the bottom surface of the mat foundation is assumed to be fixed.

2.6 Loading Conditions

(1) Input Earthquake Motion

The criteria for the selection of horizontal input earthquake motions specified in Reference (11), Methodology for Performance Evaluation of Buildings Using Time-History Response Analysis (Japan Building Center), for high-rise buildings are outlined below.

(1) Earthquake motions having the acceleration response spectrum specified for the engineering bedrock in Notification No. 4, Item (a) and generated by appropriately considering the amplification effects of the surface soil layers at the construction site (hereinafter referred to as "Notification Waves") shall be used as the design input earthquake motions.

In this case, three or more earthquake records shall be used that satisfy the requirements specified in Notification No. 4, Item (a), including duration and other relevant conditions, and that have been generated with appropriate consideration of phase characteristics.

(2) In accordance with the proviso of Notification No. 4, Item (a), when simulated earthquake motions for the construction site (hereinafter referred to as "Site-Specific Waves") are appropriately developed based on the distribution of active faults surrounding the site, fault rupture models, historical seismic activity, ground conditions, and other relevant factors, such site-specific waves may be used as the design input earthquake motions in place of the earthquake motions corresponding to extremely rare seismic events specified in the Notification Waves described above.

In this case, three or more earthquake records generated with appropriate consideration of phase characteristics and other relevant factors shall be used. When Notification Waves are used together with Site-Specific Waves, the total number of records shall be three or more.

(3) In either case described in (1) or (2) above, the following earthquake motions shall also be used as design input earthquake motions in order to verify the appropriateness of the generated earthquake records.

Specifically, from among representative recorded earthquake motions observed in the past, three or more records shall be selected appropriately, taking into consideration the characteristics of the construction site and the building.

Earthquake motions generated by scaling the maximum velocity amplitude of these records to:

- 250 mm/s shall be regarded as earthquake motions that occur rarely, and
- 500 mm/s shall be regarded as earthquake motions that occur extremely rarely.

Furthermore, the above values of maximum velocity amplitude may be multiplied by the seismic zone coefficient Z specified in Article 88, Paragraph 1 of the Enforcement Order.

4.4.1 Selection of Horizontal Input Earthquake Motions

As described above, for the performance evaluation of high-rise buildings using time-history response analysis, it is necessary to use at least three code-specified artificial ground motions, at least three site-specific ground motions, and at least three recorded earthquake ground motions. The code-specified and site-specific ground motions must be generated by the designer in accordance with the Technical Guidelines (Draft) for the Generation of Design Input Ground Motions, Reference (10).

However, the subject structure of this study is a two-story building, and the structural performance of low-rise buildings is generally evaluated using conventional static seismic force procedures.

For the dynamic nonlinear seismic response analysis conducted in this study, the El Centro NS ground motion (peak acceleration: 341.7 cm/s²; peak velocity: 33.5 cm/s) was selected from the twelve representative recorded earthquake motions shown in Figures 2.14 through 2.25 and Appendix 2, as identified by the High-Rise Building Structural Evaluation Committee in Building Letter (June 1986, Building Center of Japan) as the most reliable currently available. The earthquake motion was applied in a single horizontal direction corresponding to the weak-axis direction of the RC walls.

(2) Additional Masses of Timber Components

In the present analysis, the timber components (walls, floors, roof) are not explicitly modeled; their masses are incorporated as additional masses. The masses of the timber walls and floors are assigned to the RC walls at the floor level; the masses of the timber roof and floor systems are assigned at the roof center-of-gravity level and at the top of the RC walls. The values and locations of these additional masses are presented in Appendix 1.

2.7 Soil Springs

In this analysis, the bottom surface of the mat foundation is modeled using horizontal and rotational soil springs to account for the dynamic soil–structure interaction. The soil spring constants are determined based on the vibration admittance theory presented in Reference (5), JEAG 4601, pp. 318–321. Static soil springs are adopted in the present analysis.

(1) Assumed Soil Conditions

Soil properties adopted are based on Reference (5), JEAG 4601, p. 233 (alluvial sand layer):

- **Alluvial Sand Layer**
- Shear Wave Velocity: **VS = 150 m/s**
- Standard Penetration Test (SPT) N-Value: **N = 15**
- Poisson's Ratio: **ν = 0.4**
- Unit Weight: **ρ = 1.8 t/m^3**
- Shear Modulus $G=\rho V_S^2/g=413(\mathrm{kgf/cm^2})$
- Young's Modulus $E=2(1+\nu)G=1157(\mathrm{kgf/cm^2})$

(2) Vertical Soil Spring Constant (Self-Weight Condition)

$$K_V=G\sqrt{(A)}\times a_V/(1-\nu)$$
$$=40.5\times\sqrt{(10310\times13500)}\times2.15/(1-0.4)$$
$$\therefore K_V=1.713\times10^6(\mathrm{N/mm})$$

(3) Horizontal and Rotational Soil Spring Constants (Seismic Condition)

$$K_H=G\sqrt{(A)}\times a_H$$
$$=40.5\times\sqrt{(10310\times13500)}\times2.75$$
$$\therefore K_H=1.314\times10^6(\mathrm{N/mm})$$

- Rotational Soil Spring Constant

$$K_R=\pi GZy\times a_R/(1-\nu)$$
$$=\pi\times40.5\times(10310\times13500^2/6)\times2.3/(1-0.4)$$
$$\therefore K_H=1.527\times10^{14}(\mathrm{N\cdot mm/rad})$$

2.8 Damping Characteristics

For the material damping characteristics, Rayleigh damping (blue line in Figure 2.30) is used together with the ABAQUS *DAMPING option. The damping coefficients α and β are determined based on the eigenvalue analysis results of Chapter 3, focusing on the first natural frequency f_1 (largest participation factor; first bending mode of partition wall, red circle) and the tenth natural frequency f_{10} (relatively large participation factor; second bending mode of partition wall, yellow circle). Based on Reference (5), JEAG 4601, the damping ratio for reinforced concrete generally ranges from 0.02 to 0.05; in this analysis h = 0.03 is adopted to account for nonlinear behavior.

$$[C] = \alpha[m] + \beta[k]$$

ここに、[C]：Damping Matrix

[m]：Mass Matrix

[k]：Stiffness Matrix

α, β：Damping Coefficient

$$\alpha = h \times 2\omega_1 \times \omega_{10} / (\omega_1 + \omega_{10})$$

$$= 0.03 \times (2\pi \times 3.126) \times (2\pi \times 9.158) / ((2\pi \times 3.126) + (2\pi \times 9.158))$$

$$\therefore \alpha = 0.8786$$

$$\beta = h \times 2 / (\omega_1 + \omega_{10})$$

$$= 0.03 \times 2 / ((2\pi \times 3.126) + (2\pi \times 9.158))$$

$$\therefore \alpha = 7.7738 \times 10^{-4}$$

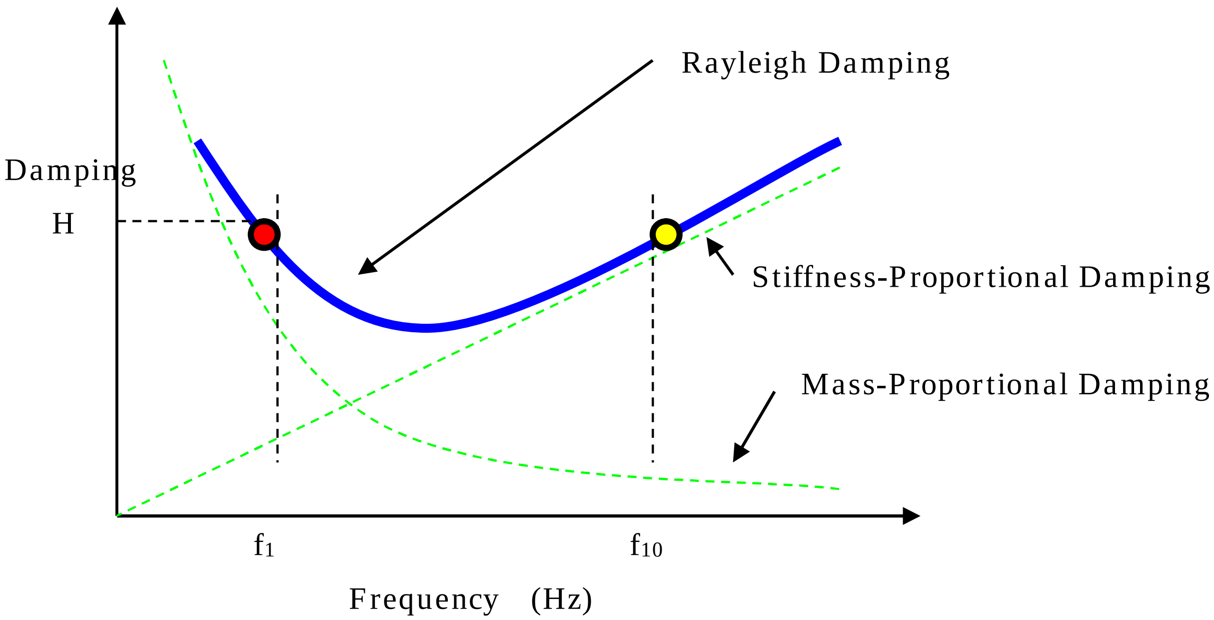


Figure 2.30 Rayleigh Damping

2.9 Analysis Method

In this study, a static self-weight analysis followed by a dynamic nonlinear seismic response analysis is performed for the two-story RC Upright Construction Method building using the direct integration method for time-history nonlinear dynamic analysis. The integration scheme used in ABAQUS/Standard is the Hilber–Hughes–Taylor (HHT) method, which is an extension of the trapezoidal rule. The HHT method is an implicit integration method, requiring the solution of the nonlinear dynamic equilibrium equations at each time increment. The nonlinear equations are solved iteratively using the Newton–Raphson method. For each time interval Δt (s) of the input earthquake motion, an implicit dynamic time integration is performed using the ABAQUS automatic time incrementation scheme, and converged solutions are obtained.

2.10 Analysis Cases

The analysis consists of one case comprising a static self-weight analysis followed by a dynamic nonlinear seismic response analysis using a single input earthquake motion record, namely the El Centro NS earthquake wave. In addition, a natural vibration analysis (eigenvalue analysis) is performed to evaluate the dynamic characteristics of the RC Upright Construction Method building.

2.11 Analysis Output

The following analytical results of the RC Upright Construction Method building are output and used as the basis for evaluating the structural integrity during earthquakes:

- Natural vibration mode shapes and animations (actual wall and slab thicknesses shown)
- Deformed shapes and animations at the time of maximum acceleration (actual wall and slab thicknesses shown)
- Acceleration contour plots at the time of maximum acceleration (actual wall and slab thicknesses shown)
- Section force contour plots at the time of maximum acceleration (actual wall and slab thicknesses shown)
- Equivalent plastic strain contour plots and animations (actual wall and slab thicknesses shown)
- Concrete cracking damage contour plots (actual wall and slab thicknesses shown)
- Acceleration time-history plots at the top of the exterior wall and at the second-floor level of the exterior wall
- Acceleration time-history plots at the top of the partition wall and at the second-floor level of the exterior wall
- Relative displacement time-history plots between the top of the exterior wall and the slab
- Relative displacement time-history plots between the top of the partition wall and the slab
- Maximum response ductility factor

Key Features of the RC Upright Construction Method

Construction Cost

"We want to create a building with outstanding architectural design. However, we also want to keep construction costs under control."

This is a common objective shared by many property owners considering rental housing investments.

The RC Upright Construction Method was developed to achieve exactly that goal.

The construction examples completed to date have been delivered at costs comparable to those of medium-grade timber residential buildings. Furthermore, through continued improvements in construction methods, engineering development, and project execution processes, we aim to achieve even greater cost reductions in the future.

At the same time, higher specifications and enhanced features can be incorporated through optional upgrades. By working closely with each owner's business plan and investment strategy, we propose attractive, high-value buildings tailored to their specific needs.

Sound Insulation

One of the most common problems in apartment buildings is noise-related disputes.

To protect the privacy and comfort of residents, the RC Upright Construction Method incorporates a highly effective sound-insulation structure into the building itself.

The 300 mm thick reinforced concrete walls provide excellent sound insulation performance, significantly reducing the transmission of noise between neighboring units.

This contributes to a more comfortable living environment and helps residents build positive relationships with their neighbors. In turn, it also provides substantial benefits to property owners by enhancing tenant satisfaction and the long-term value of the property.

Flexibility

Trends are constantly changing, and the requirements placed on buildings will continue to evolve over the next 10 or 20 years. In an era of such rapid change, adaptability and ease of renovation are essential considerations.

By retaining the reinforced concrete (RC) structural walls that form the building's permanent foundation, while allowing the other elements—such as walls, floors, and roofs—to be redesigned and renewed in accordance with changing lifestyles and architectural trends, the building can continue to meet market demands and attract occupants for generations to come.

Impact of Full Renovation Capability on Asset Value

In rental apartment buildings, once residents are living in their units, renovation work can be difficult to undertake. Property owners cannot simply require tenants to move out in order to carry out construction.
The RC Upright Construction Method is designed with a high degree of adaptability, allowing for partial renovations and modifications without requiring the entire building to be rebuilt. This flexibility enables rental housing to respond effectively to changing market demands, lifestyles, and design trends over time. As a result, it provides a sustainable and future-ready housing solution for both owners and residents.

Figure 2.1 Key Features of the RC Upright Construction Method (1/2)

Advantages of Condominium Ownership

Many property owners build rental housing with the intention of eventually passing the land and buildings on to their children as part of their estate planning.
The RC Upright Construction Method, which allows for sectional ownership, is particularly well suited to such long-term asset management strategies. By enabling ownership to be divided by unit, it provides greater flexibility for inheritance planning and asset succession, making it an attractive option for owners who wish to preserve and transfer their property to future generations.

Reduced Risk of Termite Damage

Many conventional wood-frame and 2×4 construction buildings are structurally vulnerable to termite damage. Once termites infest a building, they can seriously affect critical structural components such as load-bearing columns and structural walls.

In contrast, the primary structural framework of the RC Upright Construction Method is made of reinforced concrete (RC). Therefore, even if termite damage occurs in non-structural components, it is unlikely to compromise the building's structural integrity. This provides greater durability, safety, and long-term reliability for property owners.

Fire Safety

Buildings constructed using wood-frame, 2×4, or prefabricated construction methods are generally more susceptible to the spread of fire than reinforced concrete (RC) buildings.

In the RC Upright Construction Method, the partition walls between adjacent dwelling units as well as the exterior walls are constructed with 30 cm thick reinforced concrete walls. This robust fire-resistant design is expected to provide superior fire protection performance, even compared with conventional RC buildings, helping to reduce the risk of fire spread and enhance the safety of residents and property owners alike.

Seismic Resistance

The RC Upright Construction Method utilizes a combination of heavy-duty reinforced concrete (RC) shear walls and lightweight 2×4 walls, resulting in a structure that is lighter than conventional RC buildings while maintaining excellent structural performance. (Required soil bearing capacity: 5 t/m^2).

In an actual completed project, a dynamic seismic performance evaluation was conducted. The results demonstrated that the building could withstand the Nōbi Earthquake (Magnitude 7.2) scenario anticipated by the City of Nagoya.

This advanced housing system offers significant peace of mind not only for property owners but also for tenants, who increasingly value disaster-resistant housing. Designed to address the challenges posed by natural disasters while maintaining flexibility and long-term asset value, the RC Upright Construction Method represents a next-generation approach to residential construction.

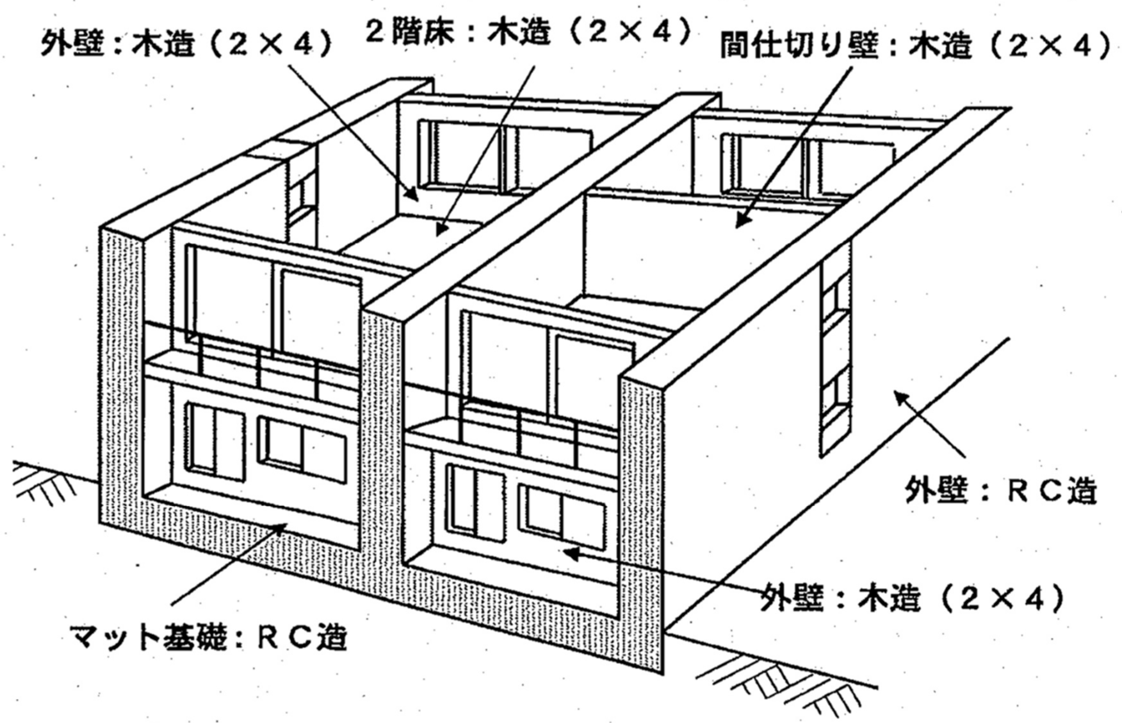
We are proud to offer this innovative and forward-looking building solution.

Design

In recent years, interest in residential design has grown significantly, to the point where it is frequently featured in television programs and magazines. We are now living in an era in which people seek to express their lifestyles and personal values through the homes they choose to live in.
To meet the expectations of these increasingly design-conscious residents, we aim to continuously introduce innovative architectural styles and living environments through collaboration with a wide range of designers. By combining flexibility, functionality, and outstanding design, we strive to create housing that remains attractive and relevant as lifestyles and trends continue to evolve.

Figure 2.1 Key Features of the RC Upright Construction Method (2/2)

Figure 2.1 RC Upright Construction Method Project Example
(Takashima Apartment Building)



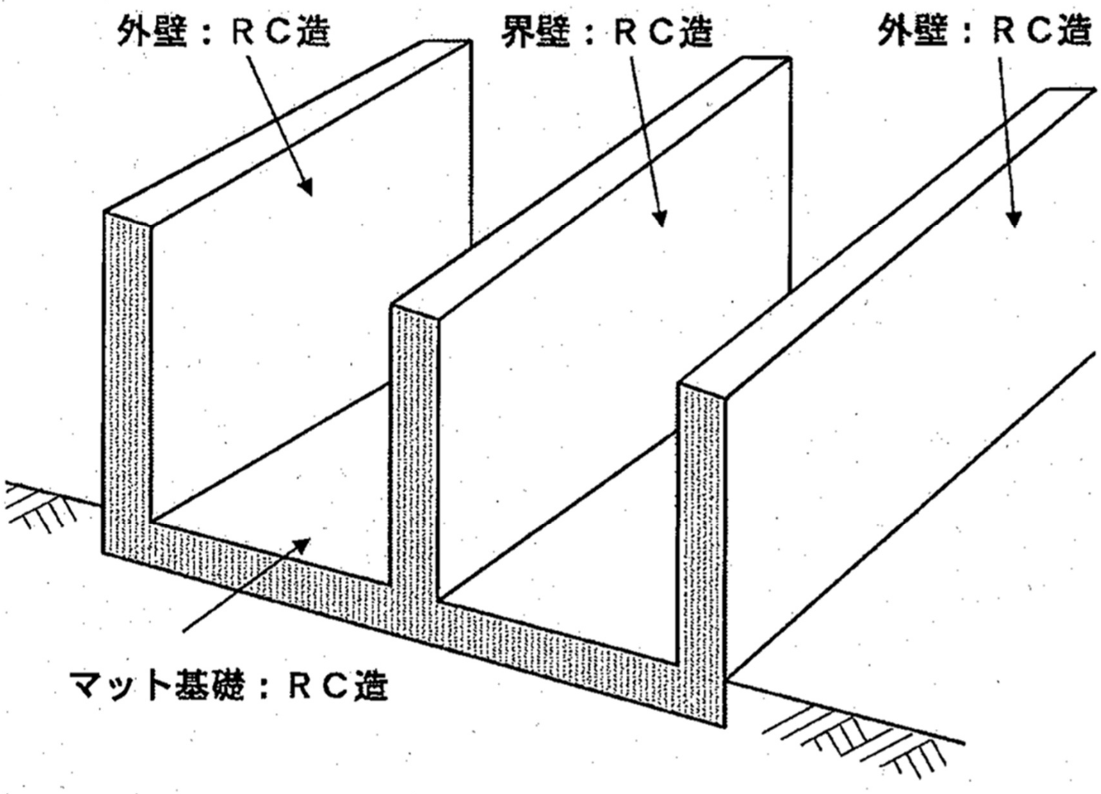


Figure 2.2 General Overview of a Two-Story Building Using the RC Upright Construction Method

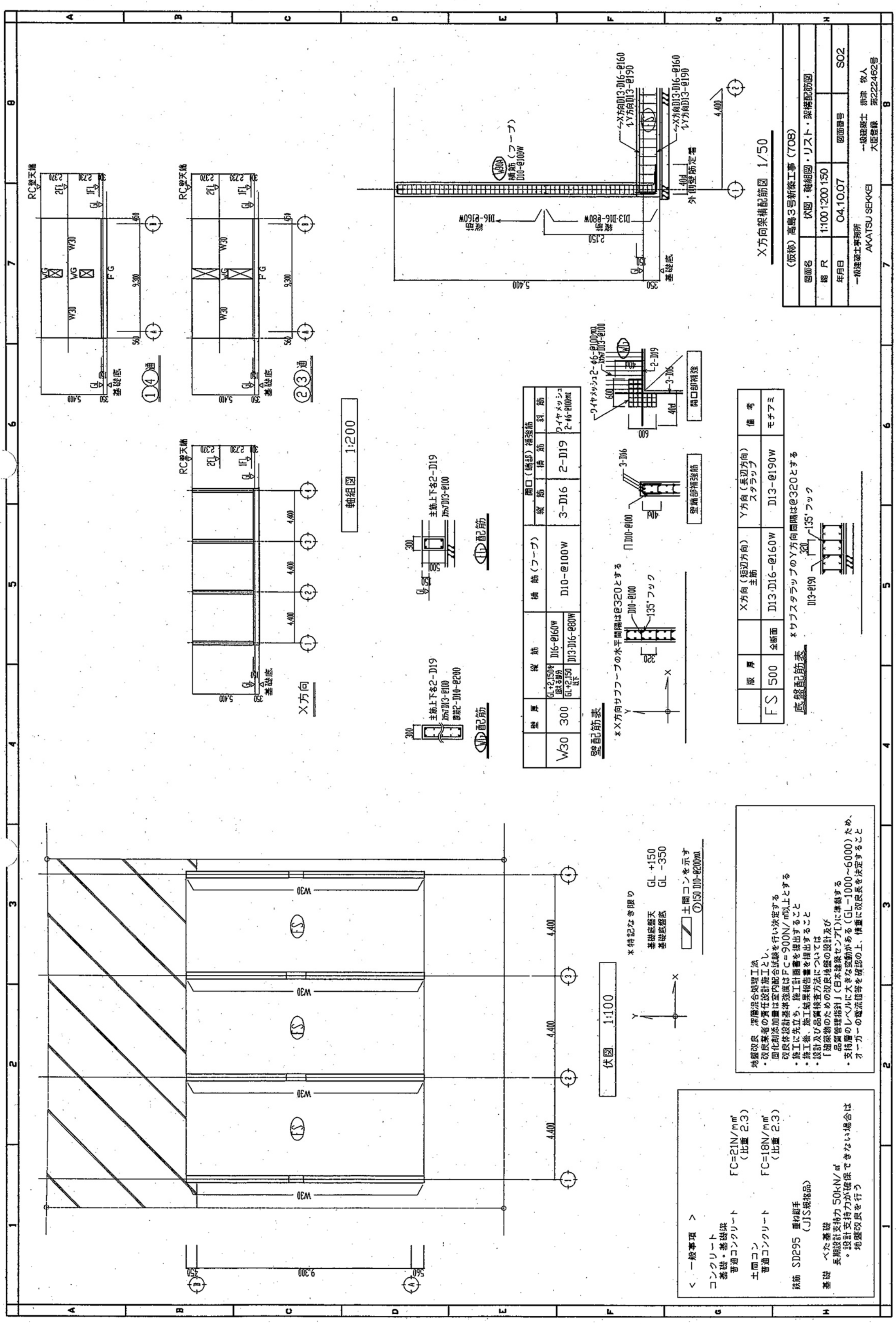


Figure 2.3 Drawings of the Subject Building for Seismic Analysis
(Two-Story RC Upright Building)

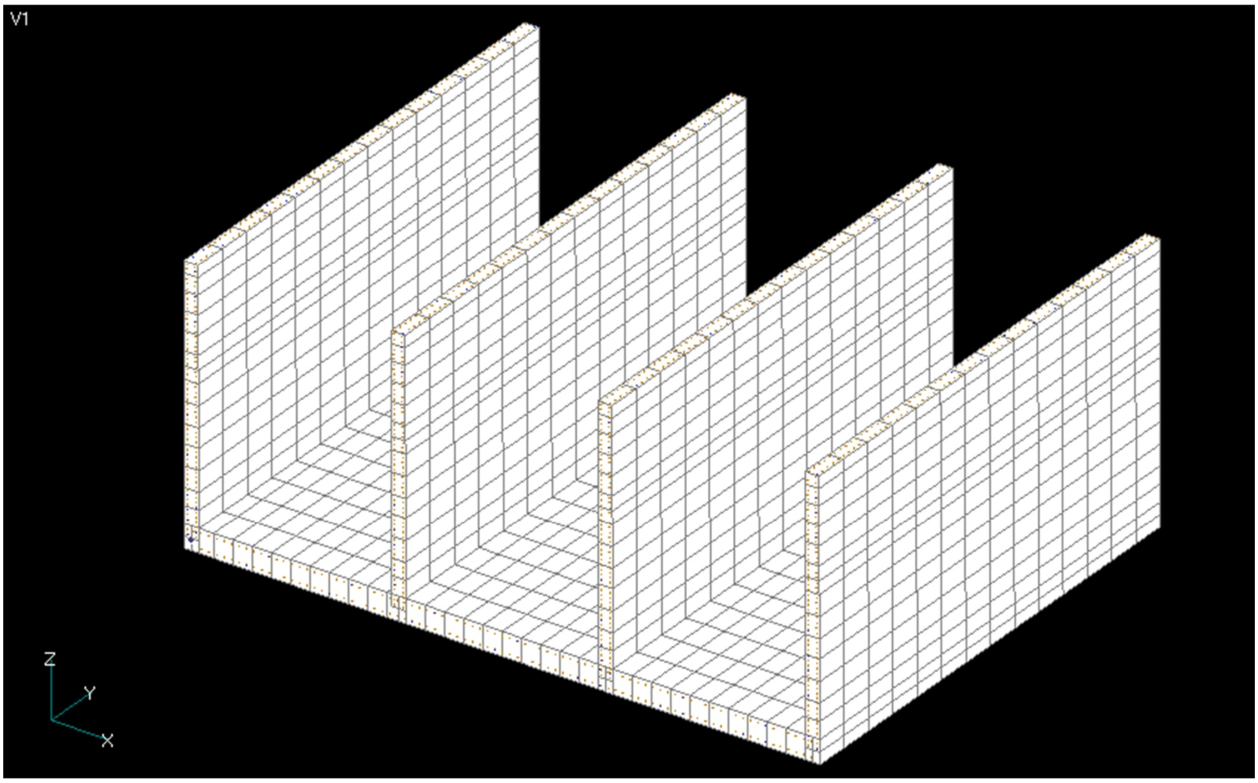


Figure 2.4 Analytical Model (Overall View with Wall and Slab Thicknesses Indicated)

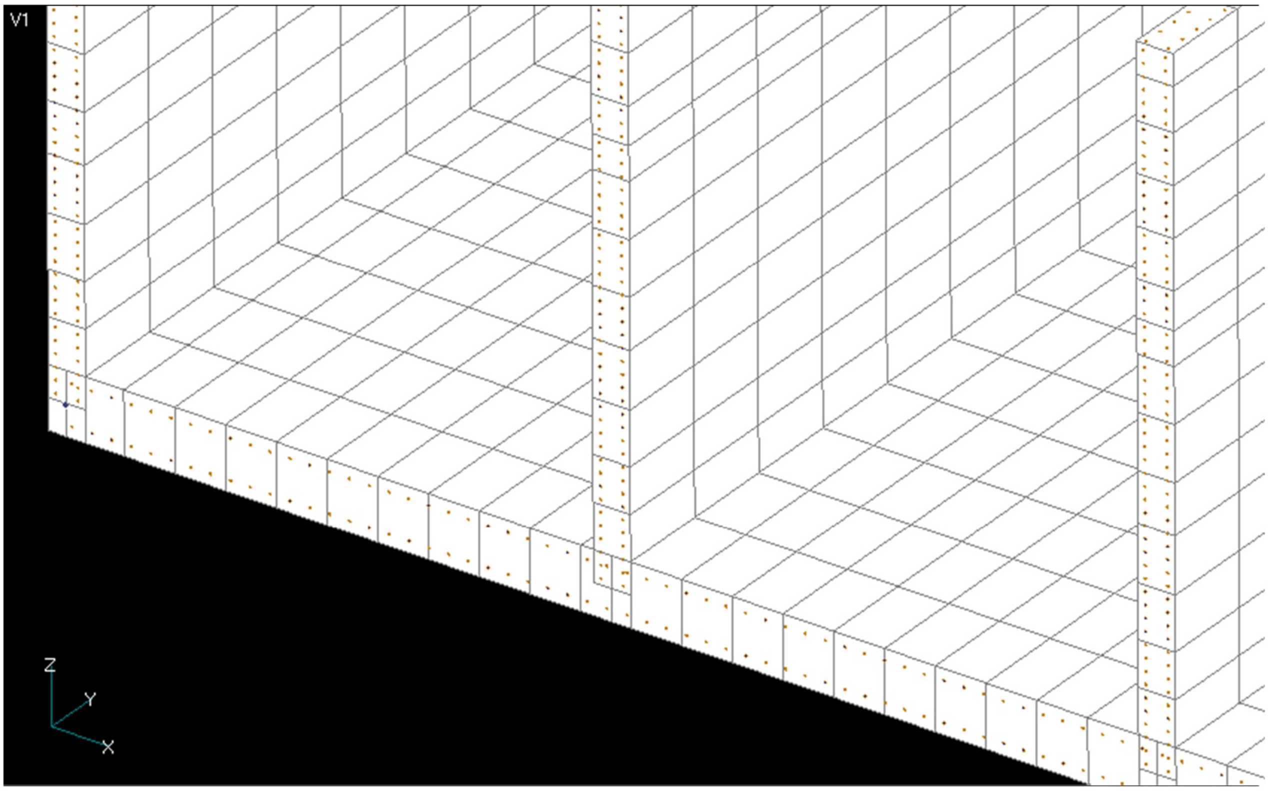


Figure 2.5 Analytical Model (Enlarged View with Wall and Slab Thicknesses Indicated)

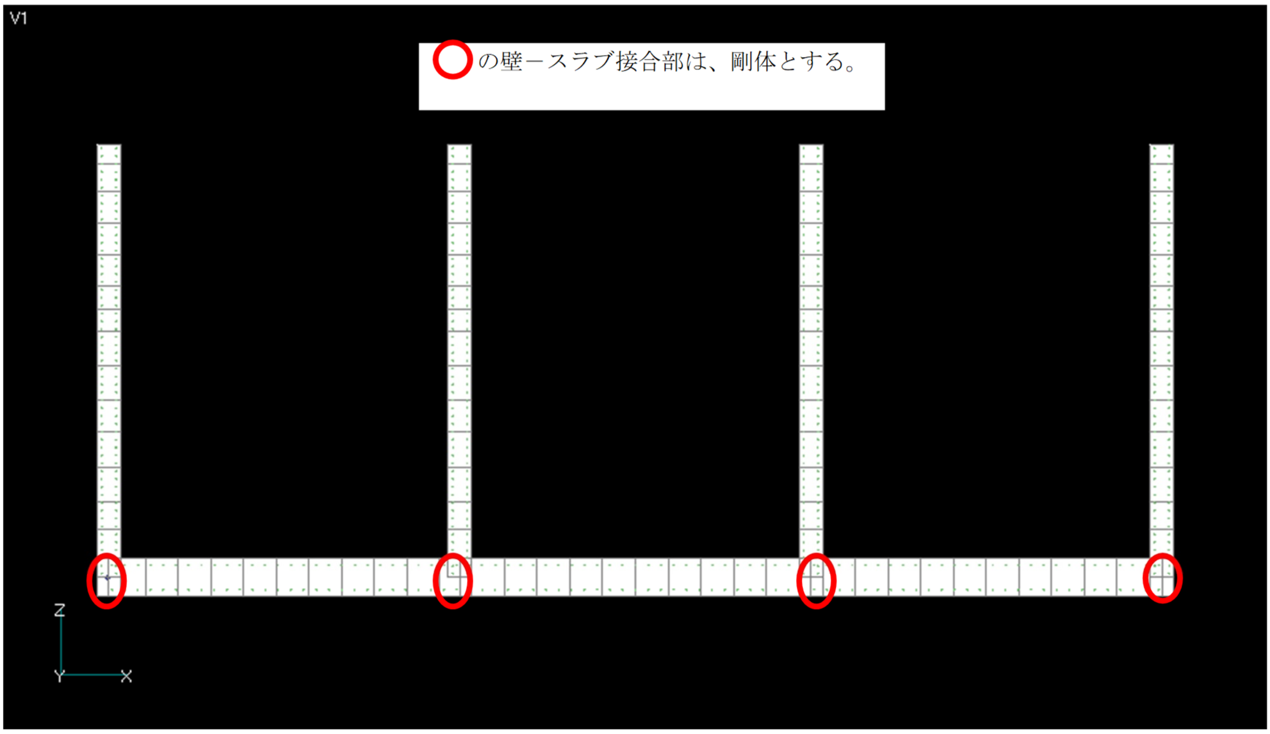


Figure 2.6 Analytical Model (Front View; Wall–Slab Connections Marked with Red Circles Are Modeled as Rigid Joints)

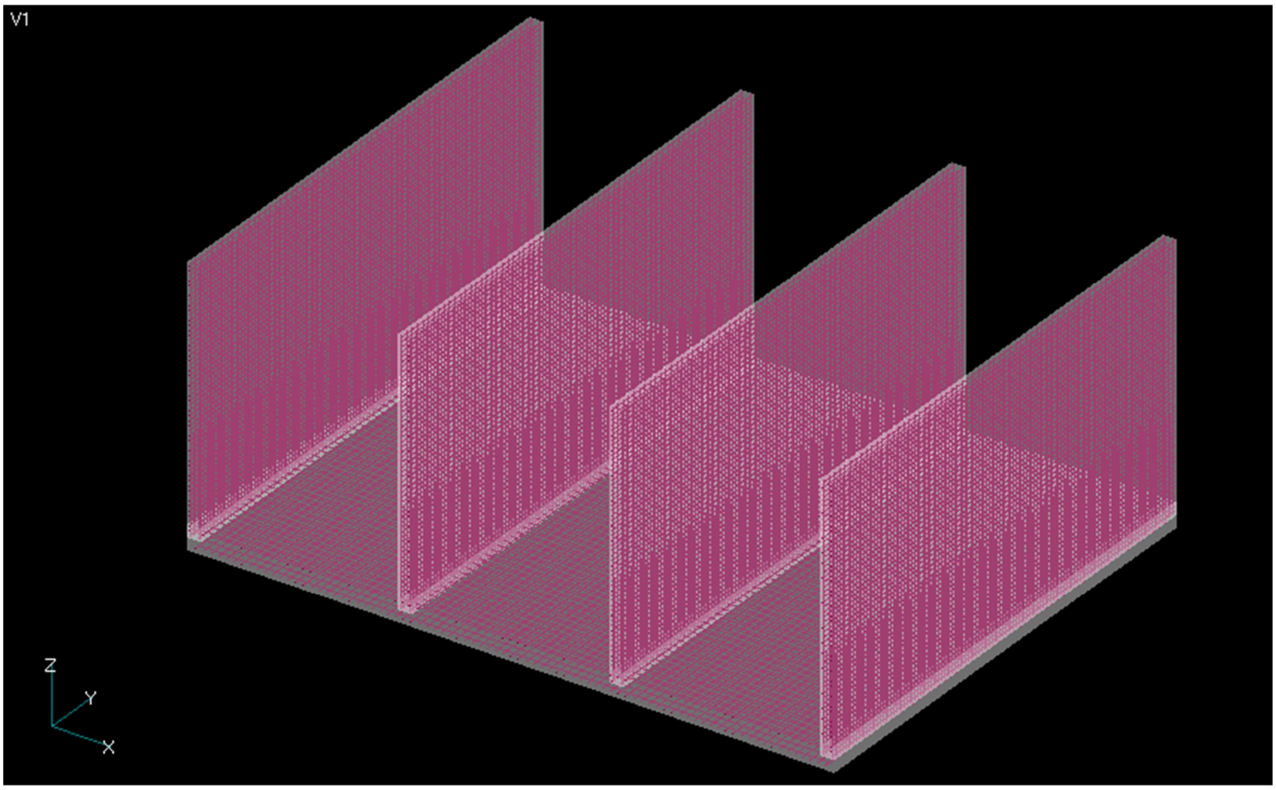


Figure 2.7 Analytical Model (Overall View with Reinforcing Bars Displayed Transparently)

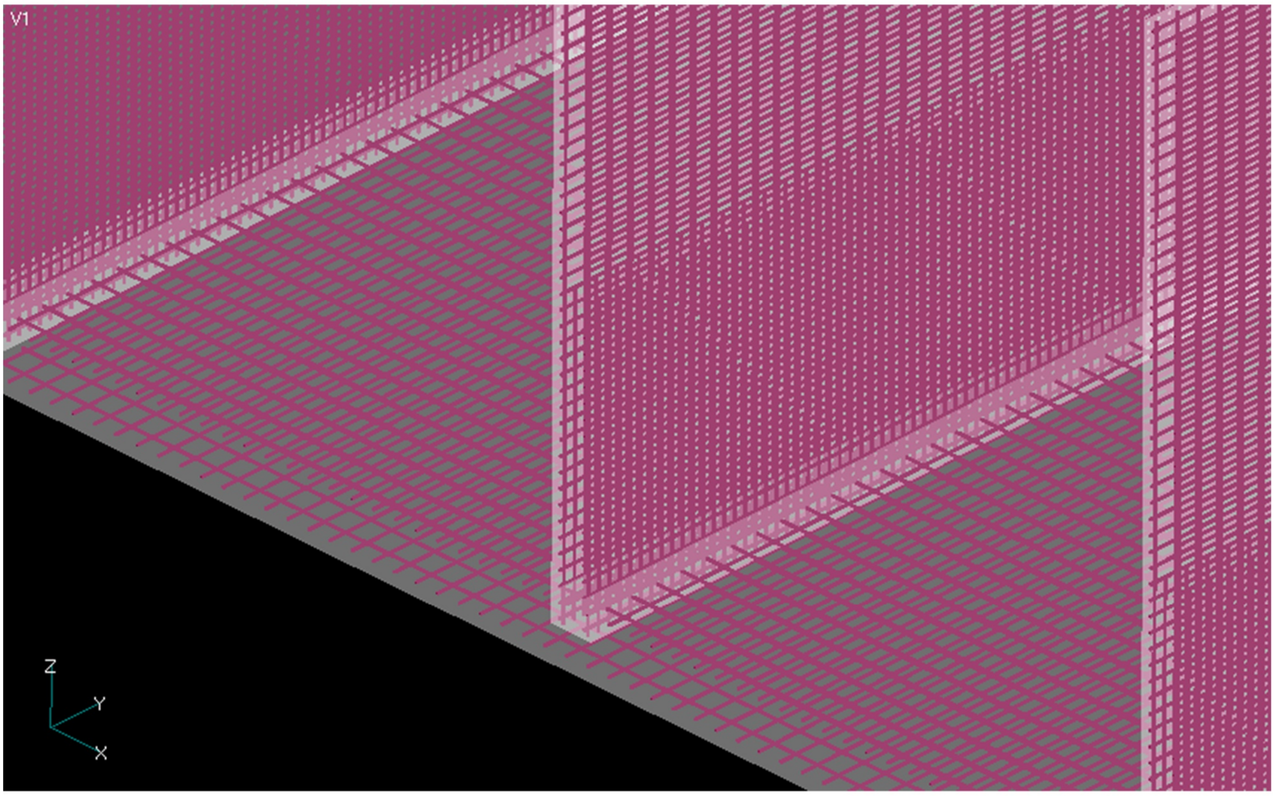


Figure 2.8 Analytical Model (Detailed View with Reinforcing Bars Displayed Transparently)

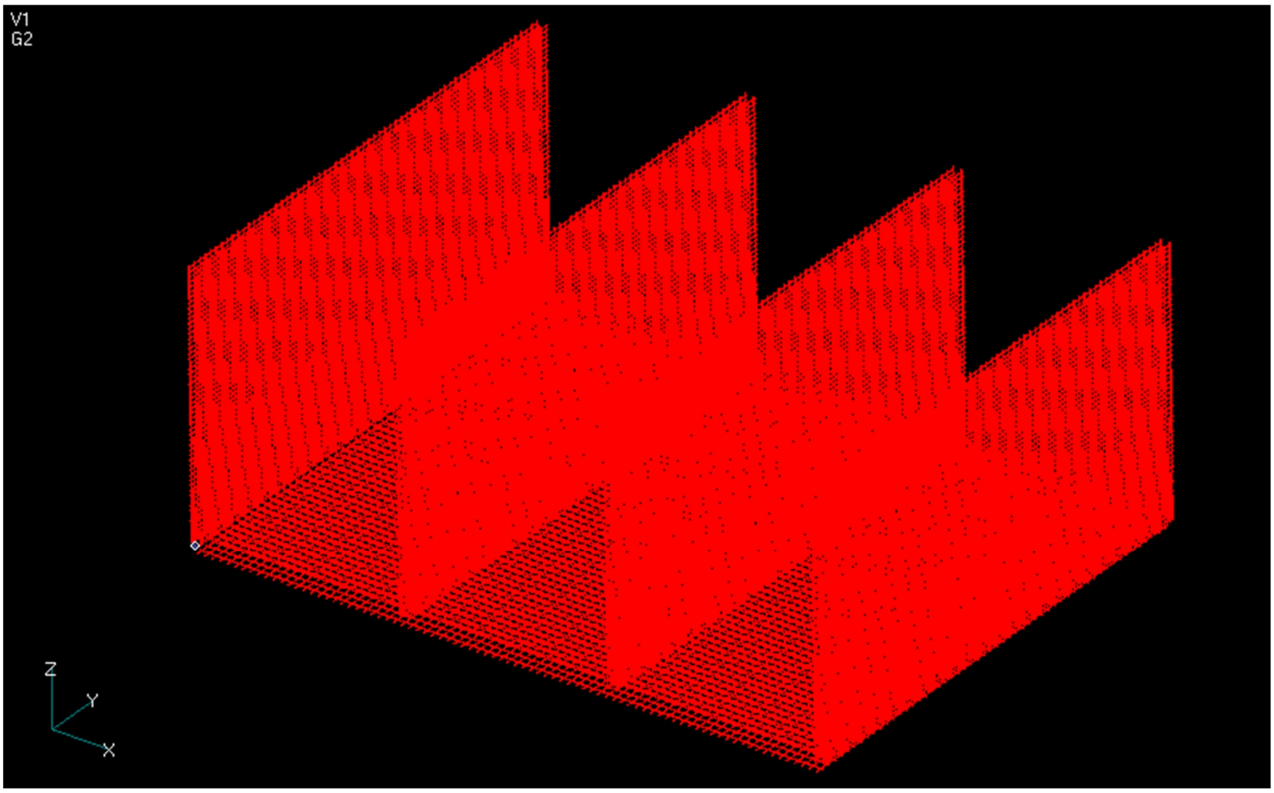


Figure 2.9 Analytical Model (Overall View Showing Reinforcing Bars Only)

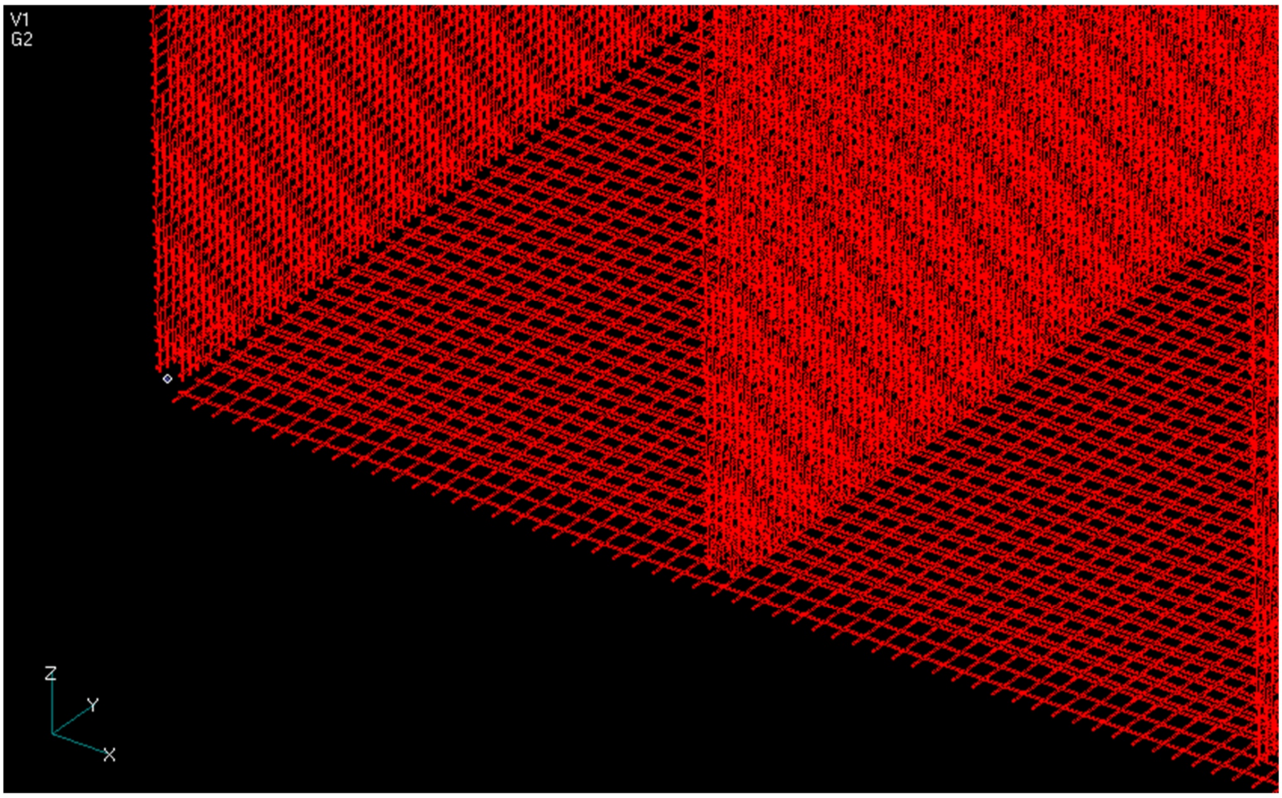


Figure 2.10 Analytical Model (Enlarged View Showing Reinforcing Bars Only)

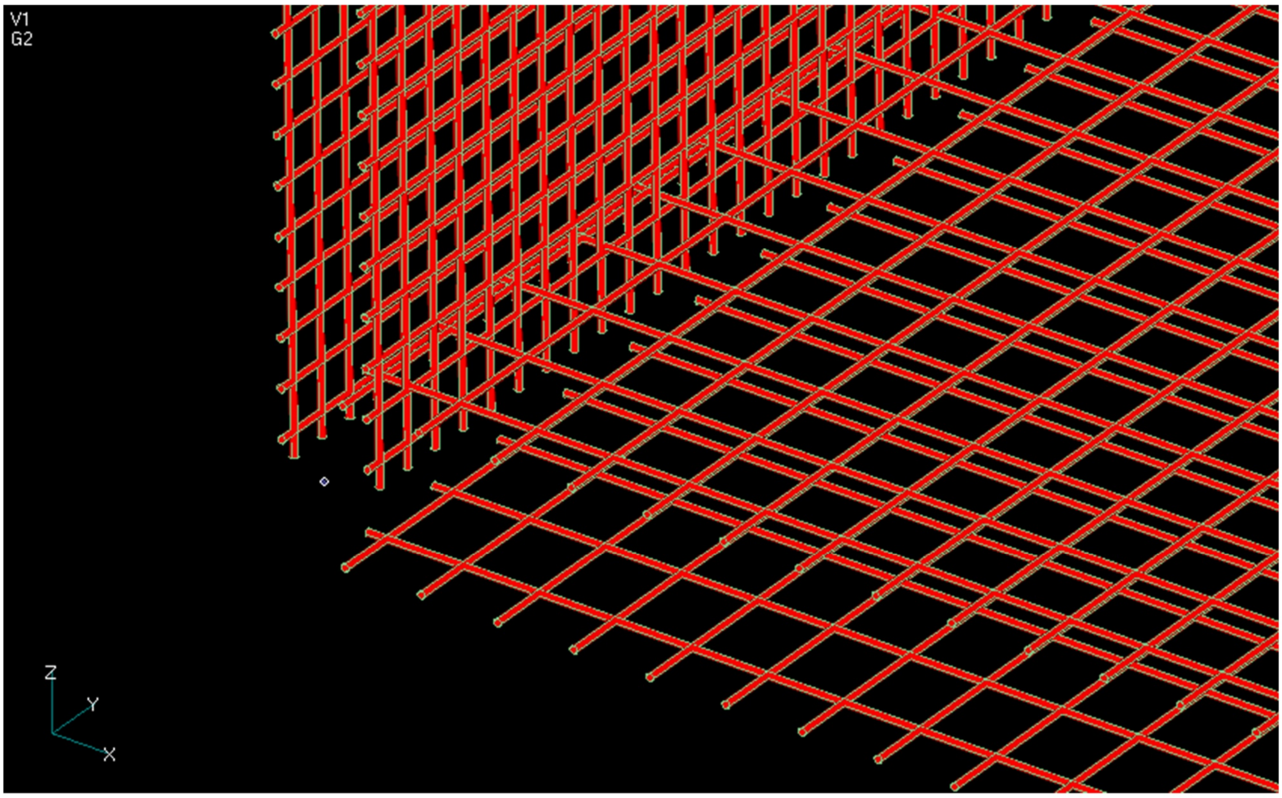


Figure 2.11 Analytical Model (Enlarged View Showing Reinforcing Bars Only)

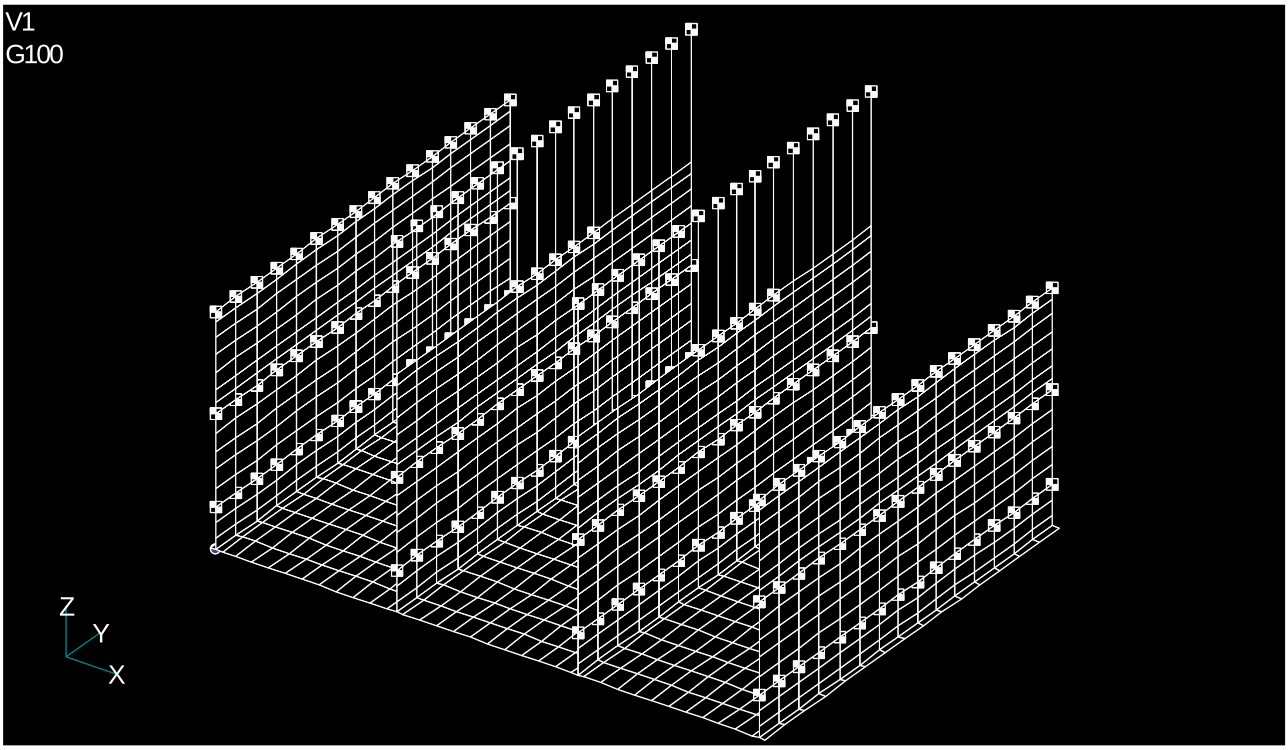


Figure 2.12 Analytical Model (Overall View Showing the Shell Element Model and Applied Additional Masses)

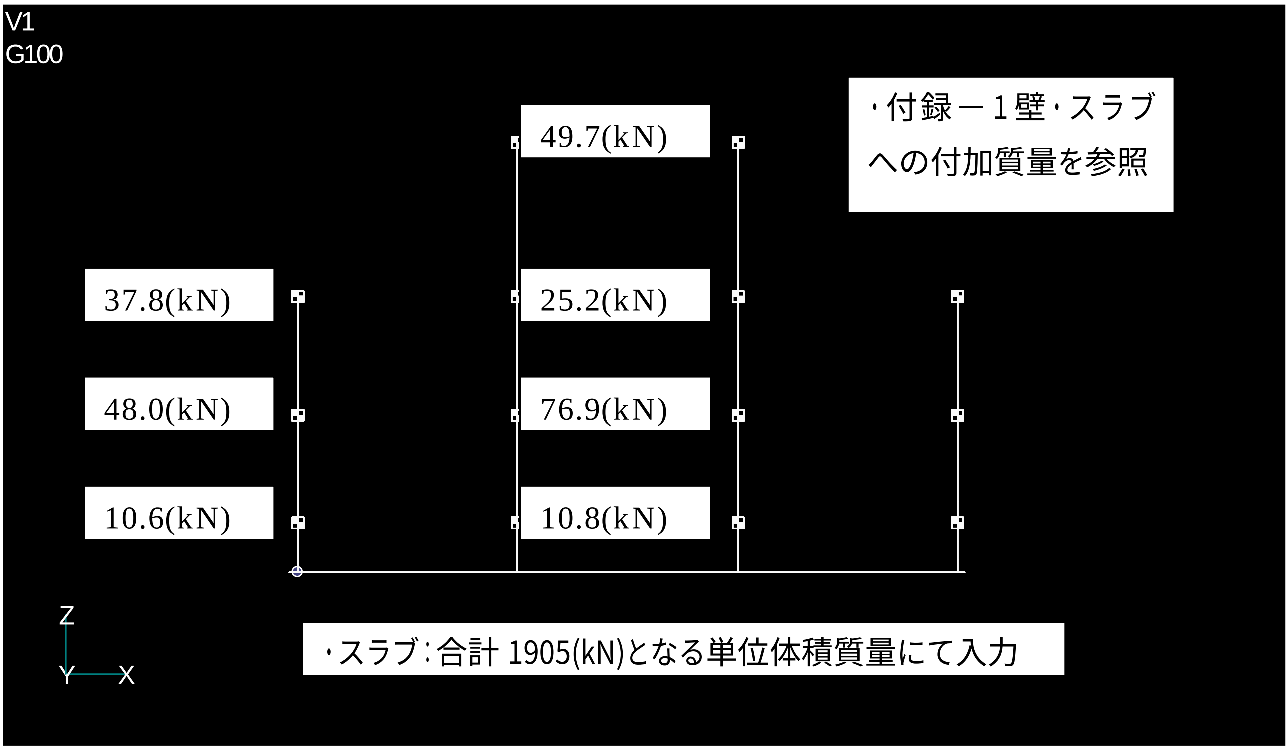


Figure 2.13 Analytical Model (Front View Showing the Shell Element Model and Applied Additional Masses)

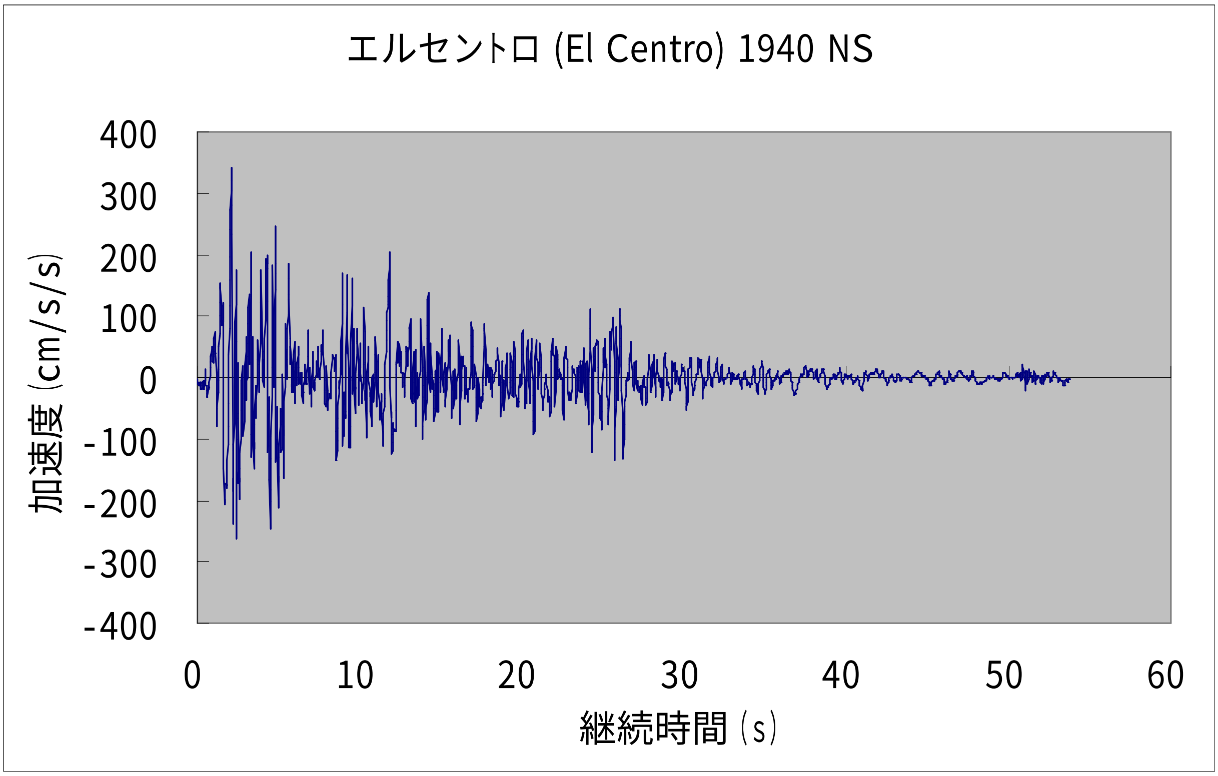


Figure 2.14 El Centro NS Ground Motion Record (Peak Ground Acceleration = 341.70 cm/s²)

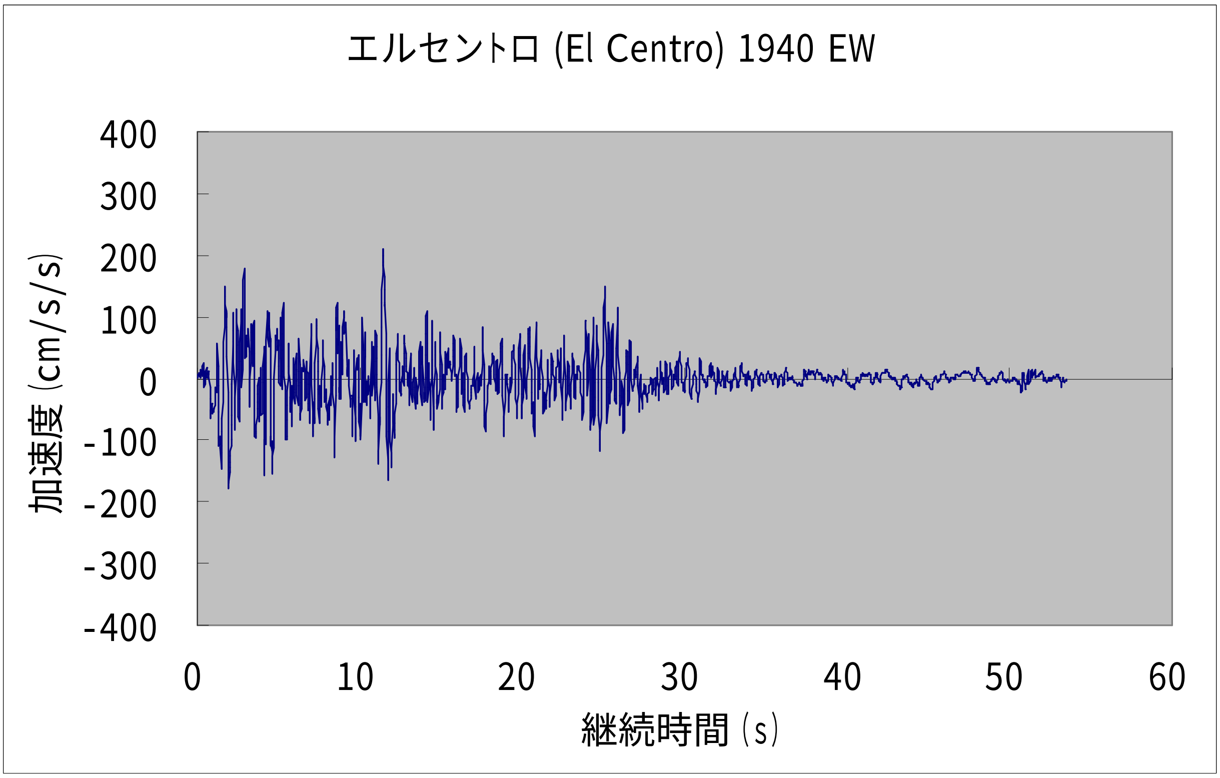


Figure 2.15 El Centro EW Ground Motion Record (Peak Ground Acceleration = 210.10 cm/s²)

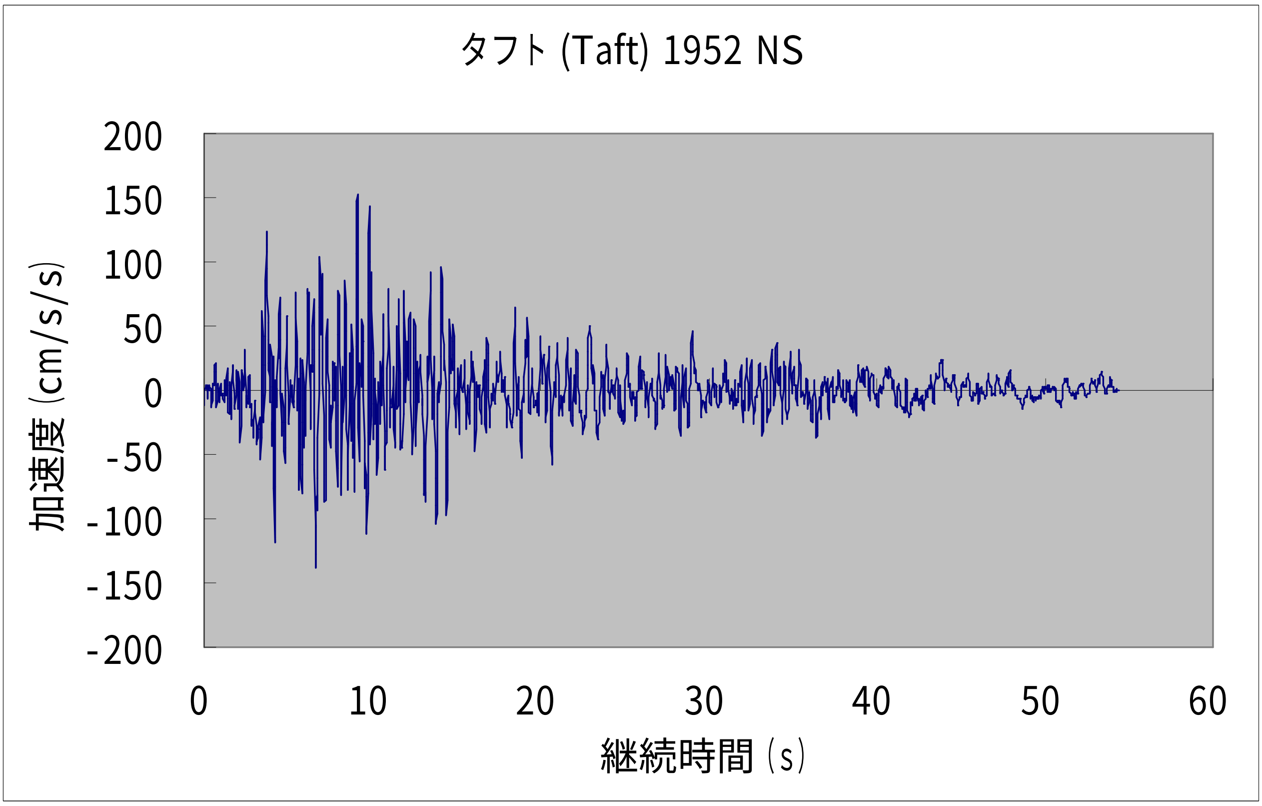


Figure 2.16 Taft NS Ground Motion Record (Peak Ground Acceleration = 152.70 cm/s²)

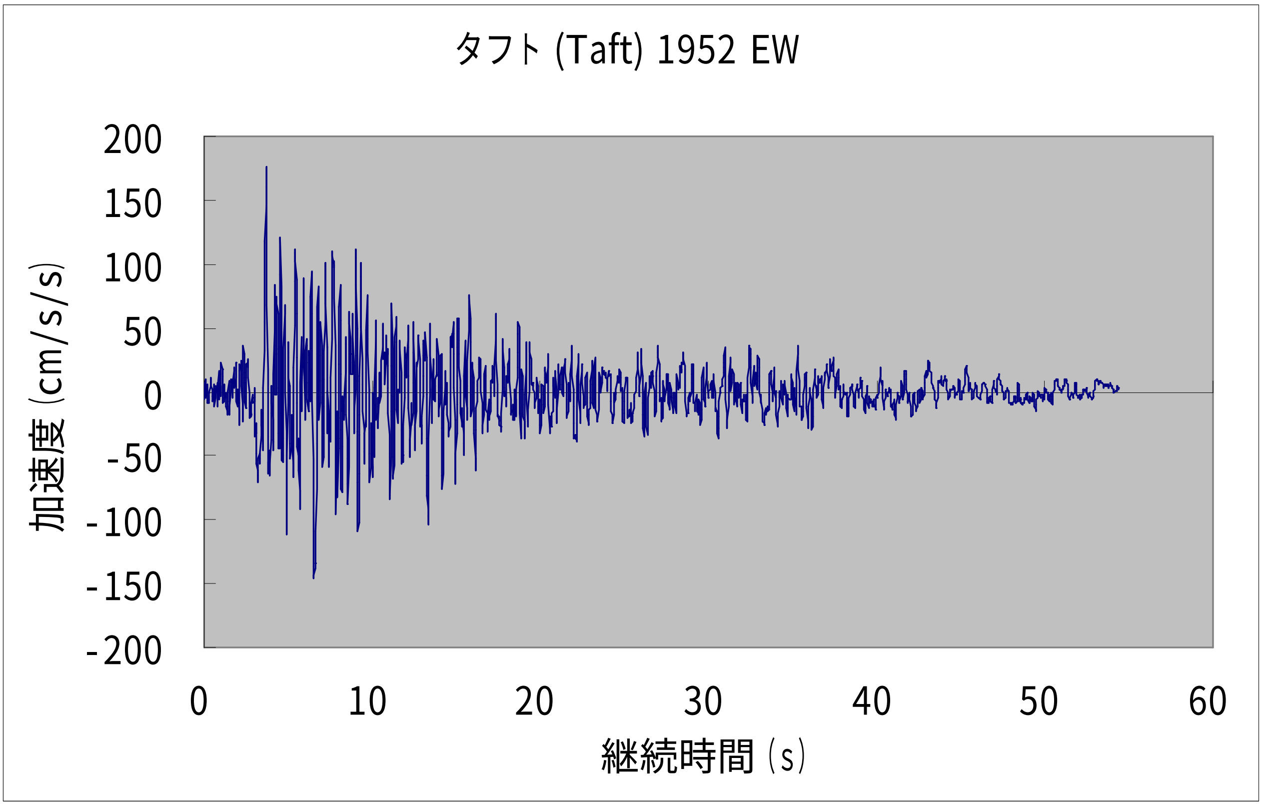


Figure 2.17 Taft EW Ground Motion Record (Peak Ground Acceleration = 175.90 cm/s²)

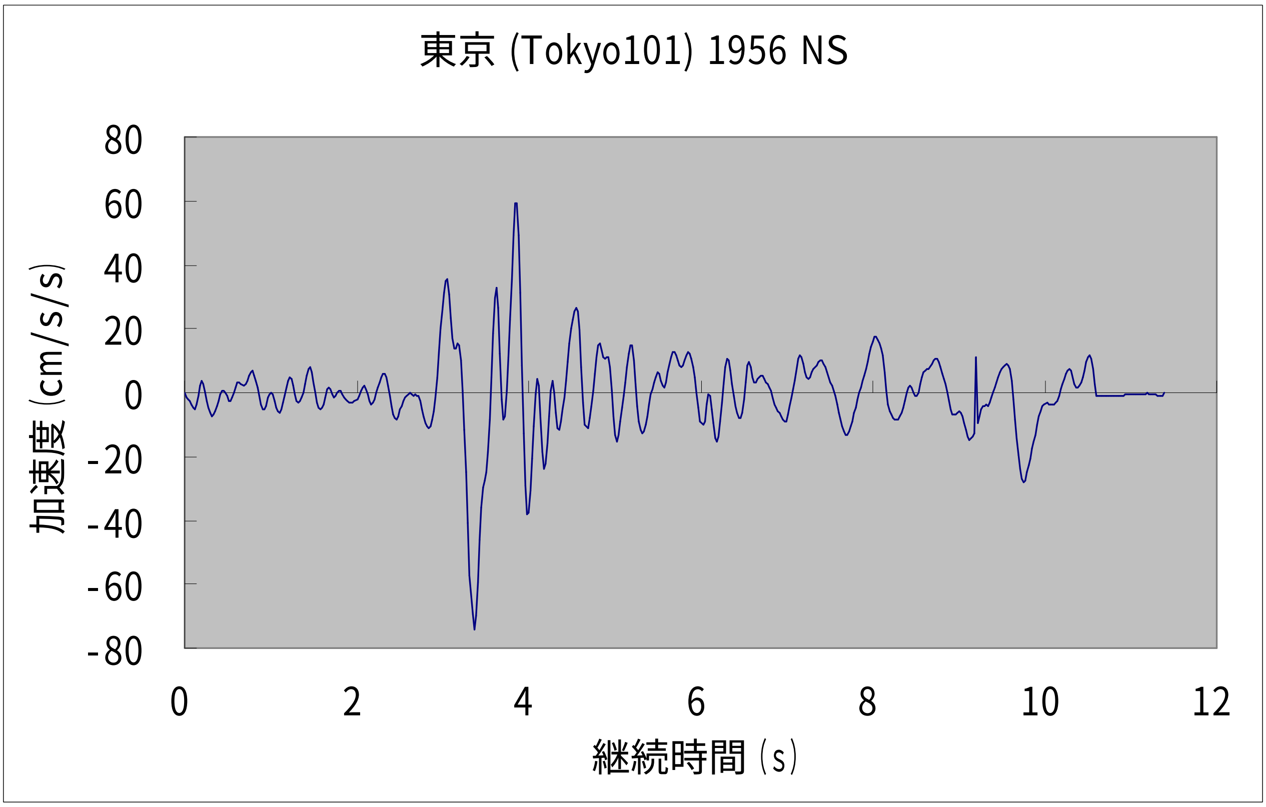


Figure 2.18 Tokyo NS Ground Motion Record (Peak Ground Acceleration = 74.00 cm/s²)

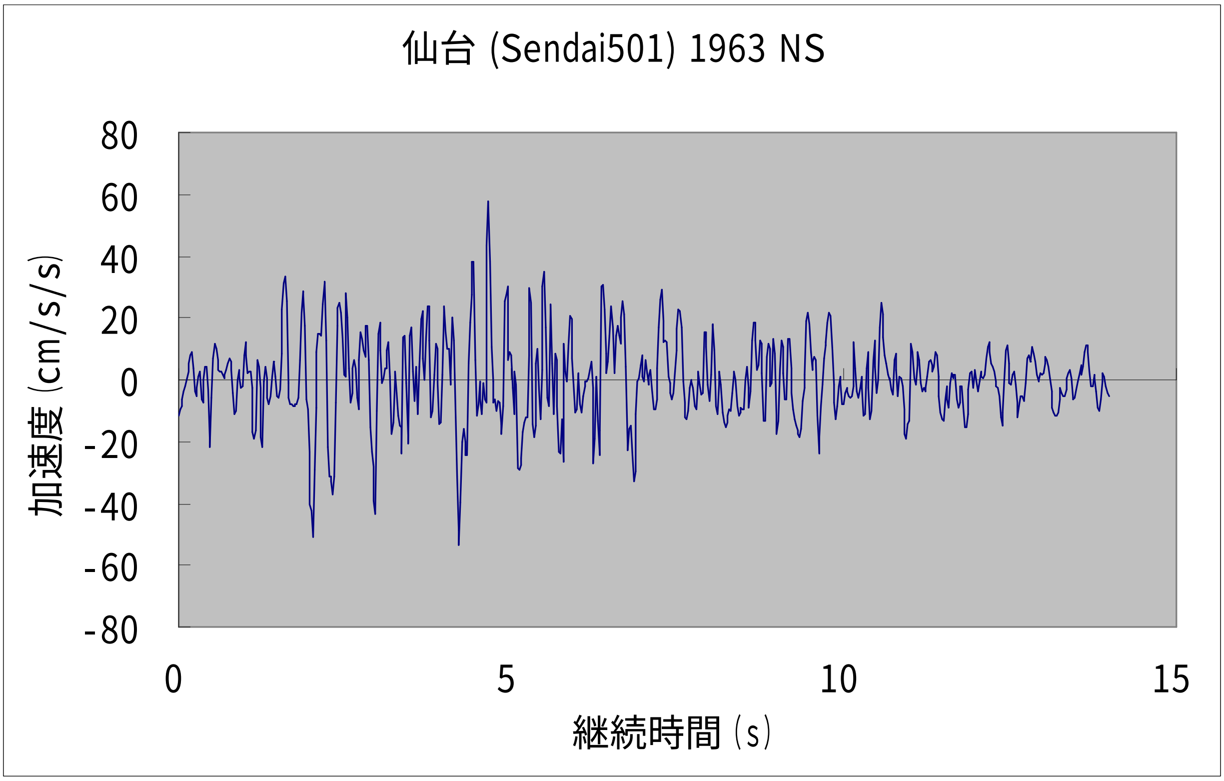


Figure 2.19 Sendai NS Ground Motion Record (Peak Ground Acceleration = 57.50 cm/s²)

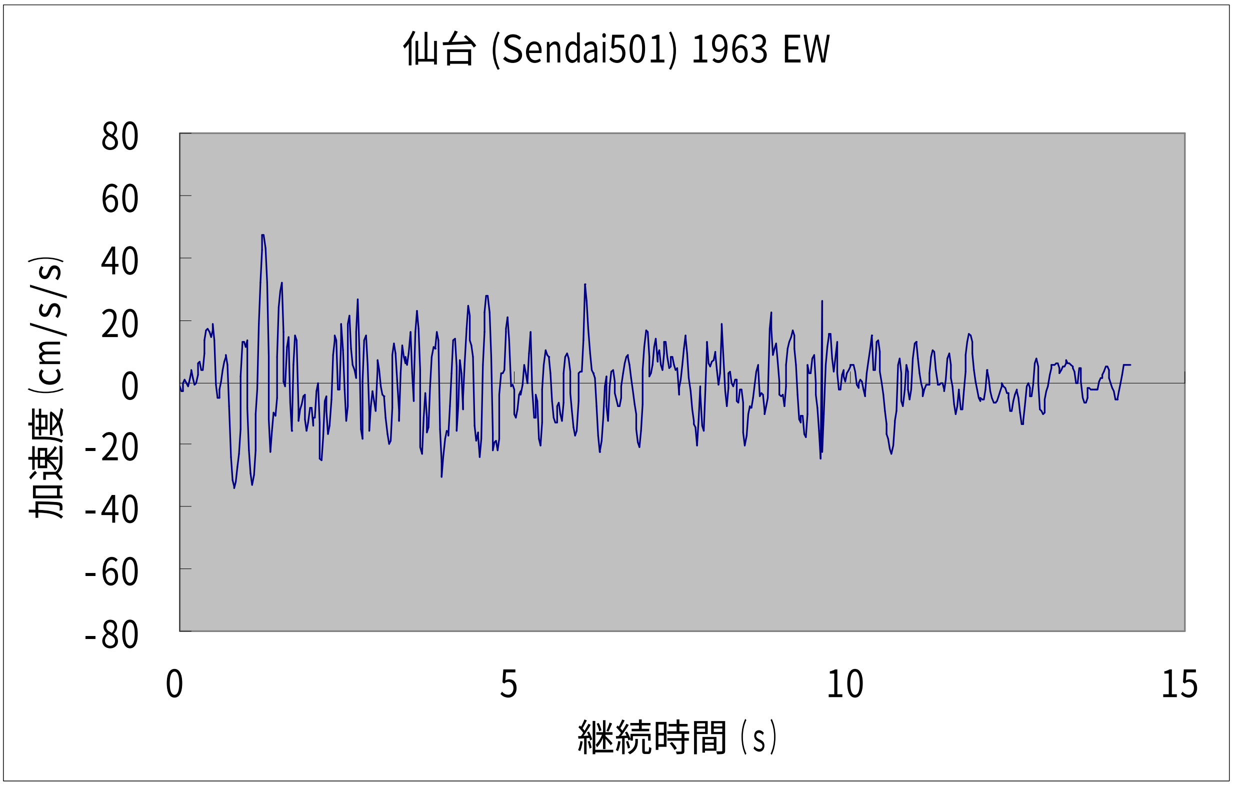


Figure 2.20 Sendai EW Ground Motion Record (Peak Ground Acceleration = 47.50 cm/s²)

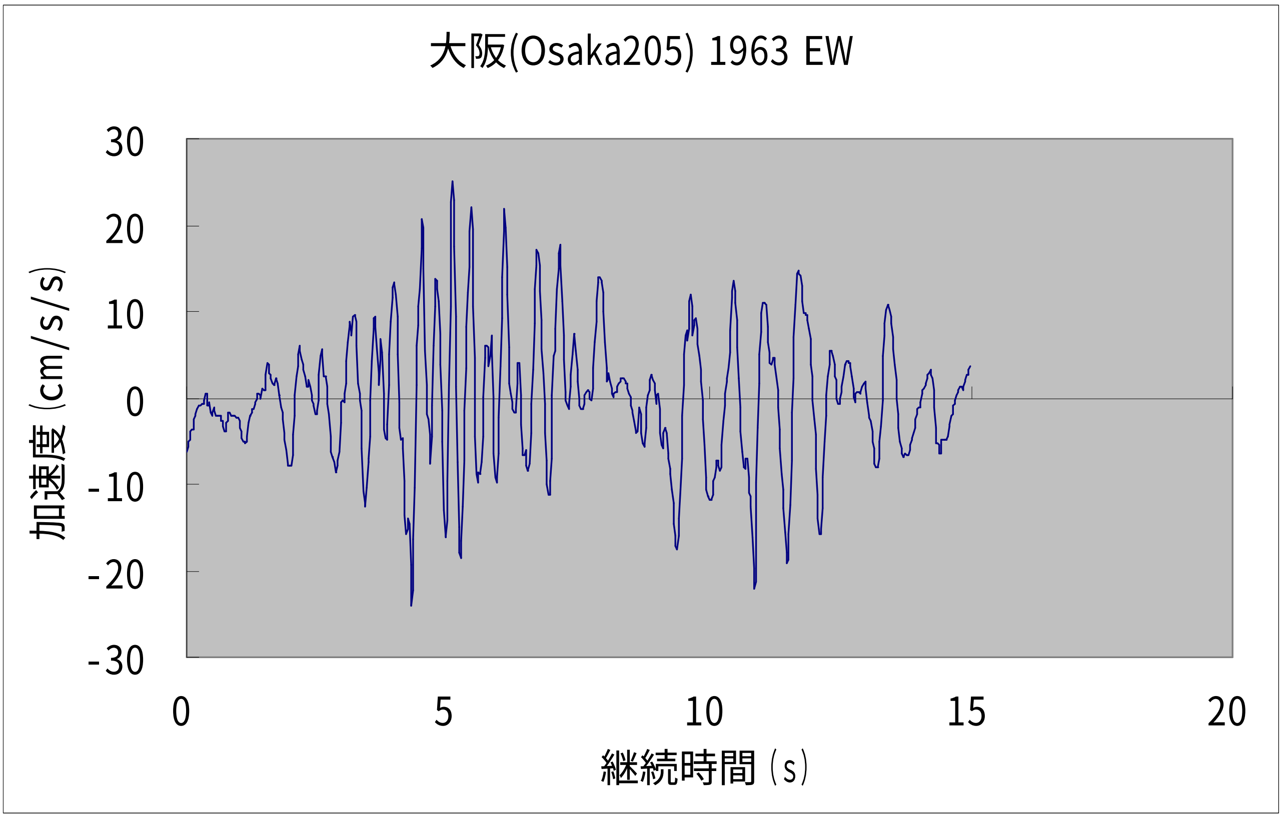


Figure 2.21 Osaka EW Ground Motion Record (Peak Ground Acceleration = 25.00 cm/s²)

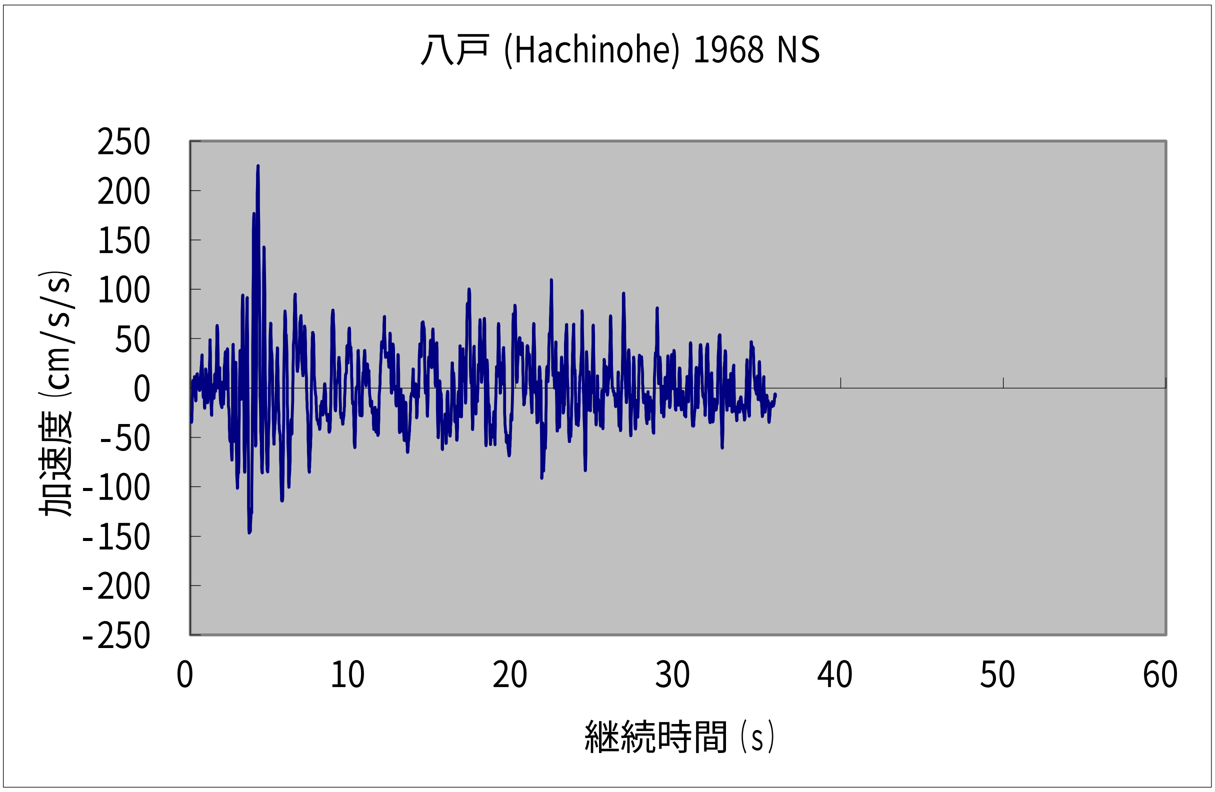


Figure 2.22 Hachinohe NS Ground Motion Record (Peak Ground Acceleration = 225.00 cm/s²)

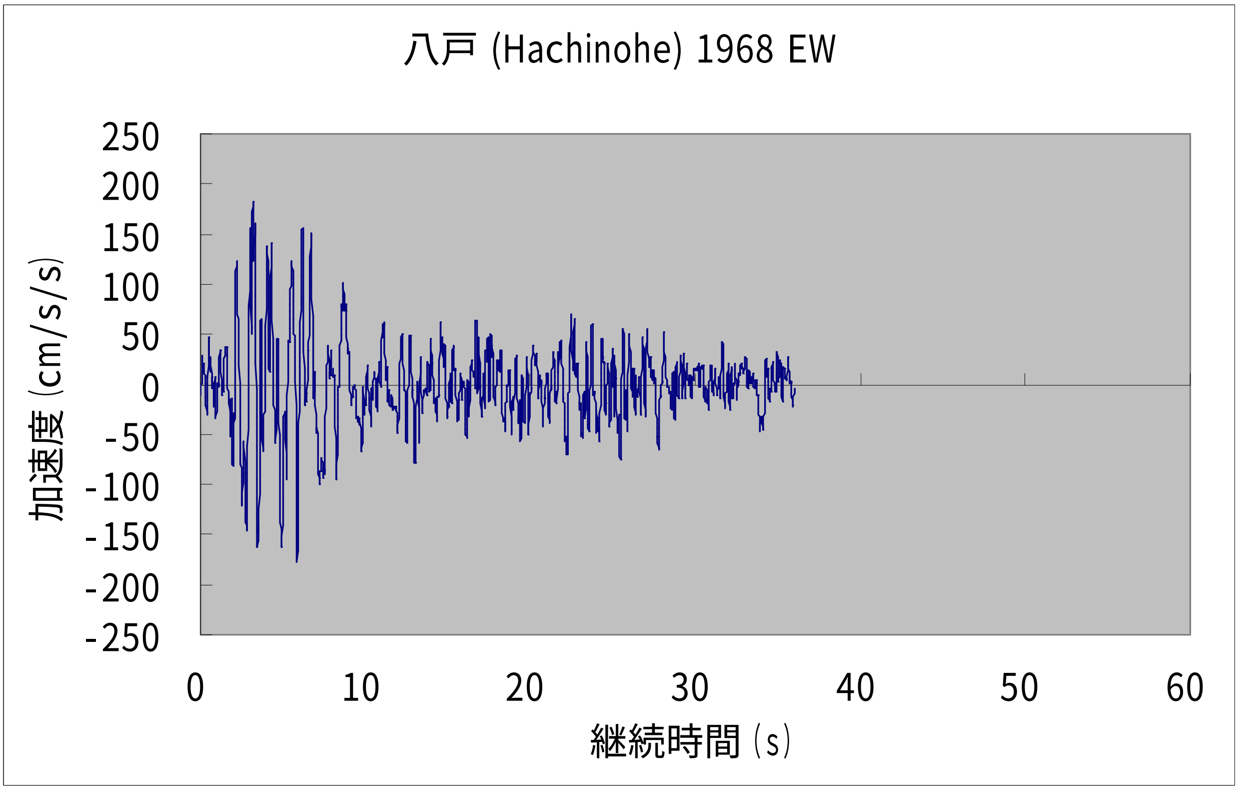


Figure 2.23 Hachinohe EW Ground Motion Record (Peak Ground Acceleration = 182.90 cm/s²)

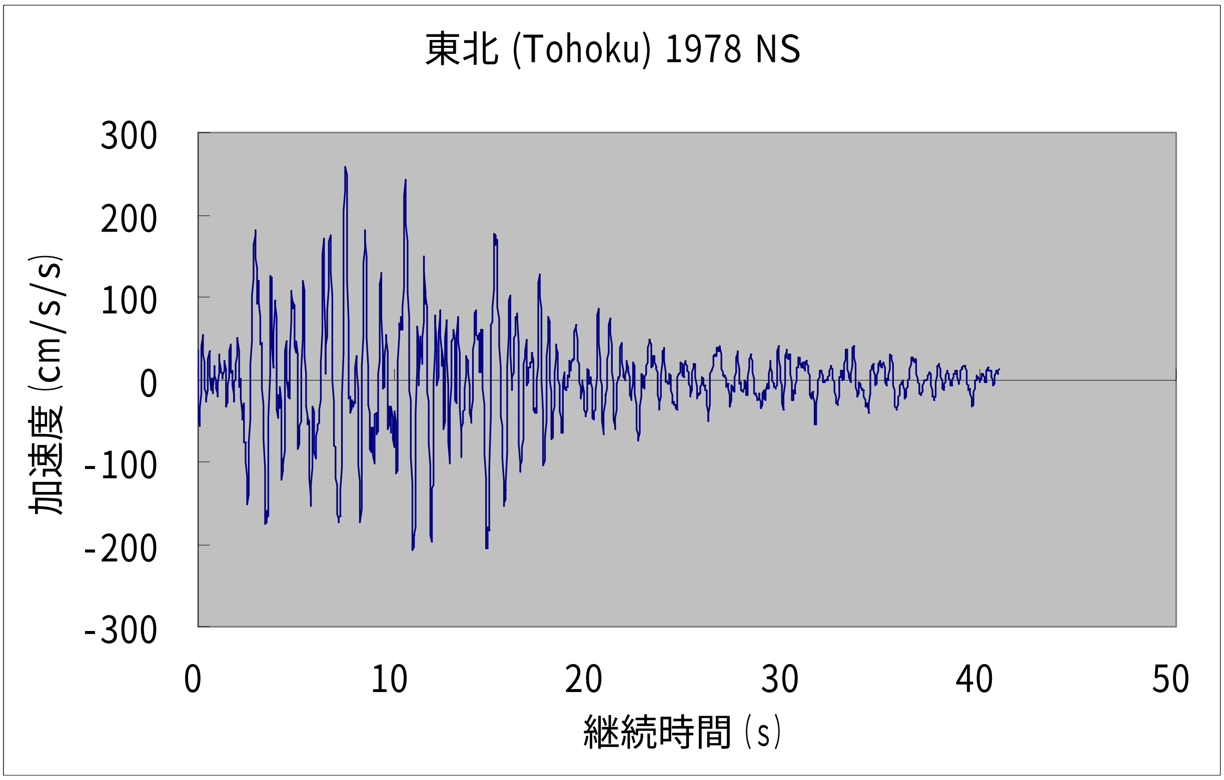


Figure 2.24 Tohoku NS Ground Motion Record (Peak Ground Acceleration = 258.20 cm/s²)

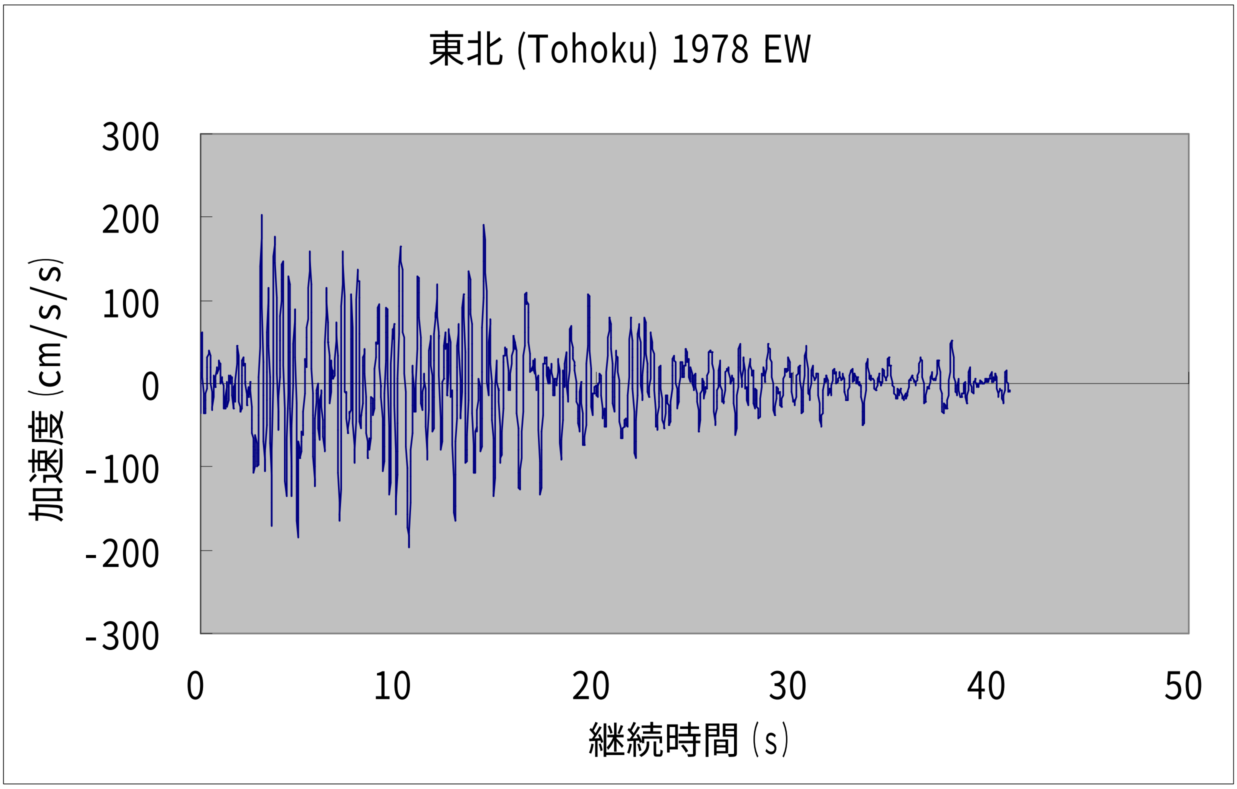


Figure 2.25 Tohoku EW Ground Motion Record (Peak Ground Acceleration = 202.50 cm/s²)

3. Analysis Results and Discussion

3.1 Eigenvalue Analysis Results

Table 3.1 presents the eigenvalue analysis results for the two-story RC Upright Construction Method building, covering the first through twelfth vibration modes. The results include natural frequencies, participation factors in the three translational directions, and the characteristics of each vibration mode.

In simple terms, the natural frequency represents the number of vibration cycles per second, and varies depending on the structural configuration and mass of the building. The participation factor is an indicator of the contribution of each vibration mode to the overall dynamic response. Note that the mode shape diagrams illustrate only the relative vibration characteristics and do not represent actual deformation magnitudes.

Incidentally, when a building is subjected to vibrations having a frequency matching one of its natural frequencies, the vibration response is significantly amplified — this phenomenon is known as resonance.

Figures 3.1 and 3.2 show the first and second natural vibration modes of the two-story RC Upright Construction Method building. These correspond to the first bending vibration modes of the partition walls, with $f_1 = f_2 = 3.126$ Hz.

Figures 3.3 and 3.4 show the third and fourth natural vibration modes, corresponding to the first bending vibration modes of the exterior walls, with $f_3 = f_4 = 4.678$ Hz.

Figure 3.5 shows the fifth natural vibration mode, corresponding to the first torsional vibration mode of the partition walls (in-phase), with $f_5 = 5.146$ Hz.

Figure 3.6 shows the sixth natural vibration mode, corresponding to the second torsional vibration mode of the partition walls (out-of-phase), with $f_6 = 5.146$ Hz.

Figures 3.7 and 3.8 show the seventh and eighth natural vibration modes, corresponding to the first torsional vibration modes of the exterior walls, with $f_7 = f_8 = 7.262$ Hz.

Focusing on the participation factors in Table 3.1, the X-direction (weak-axis direction of the partition walls and exterior walls) participation factors are large for the first through fourth vibration modes, indicating that bending vibrations in the weak-axis direction are more likely to occur.

As a validation check for the eigenvalue analysis results, the theoretical and analytical first natural frequencies are compared using a simplified model that excludes additional loads such as those from the roof. The first natural frequency f_1 of an unreinforced concrete exterior wall (height 5.4 m, thickness 0.3 m, width 10.31 m, lower end fixed) is:

$$f_1=(1.875)^2\times(\sqrt{(E\times I\times g)/(\gamma\times A)})/(2\pi L^2)$$

$$=(1.875)^2\times(\sqrt{(2.17\times10^6\times(10.31\times0.3^3/12)\times9.8)/(2.4\times10.31\times0.3)})/(2\pi 5.4^2)$$

$$\therefore\ f_1=4.95(\mathrm{Hz})$$

The first natural frequency of the RC Upright Construction Method without reinforcement, analyzed separately under the boundary condition that the slab bottom surface is fixed, was $f_1 = 4.9$ Hz for the first bending mode of the exterior wall. This is consistent with the theoretical value. Therefore, the analytical model is considered to be valid.

In addition, according to a separate field measurement survey, a vibration exciter was installed on the second floor of the building to measure the natural frequency. The measured first natural frequency was f_1 = 6.2 Hz (T_1 = 0.16 s), indicating that the building has a relatively stiff structural system. The first natural frequency obtained from the eigenvalue analysis was f_1 = 3.13 Hz (T_1 = 0.32 s), which is lower than the measured value. This is because, in this analysis, the stiffness of the roof, wooden walls, and floors was neglected; in the actual building these elements are present and increase the first natural frequency.

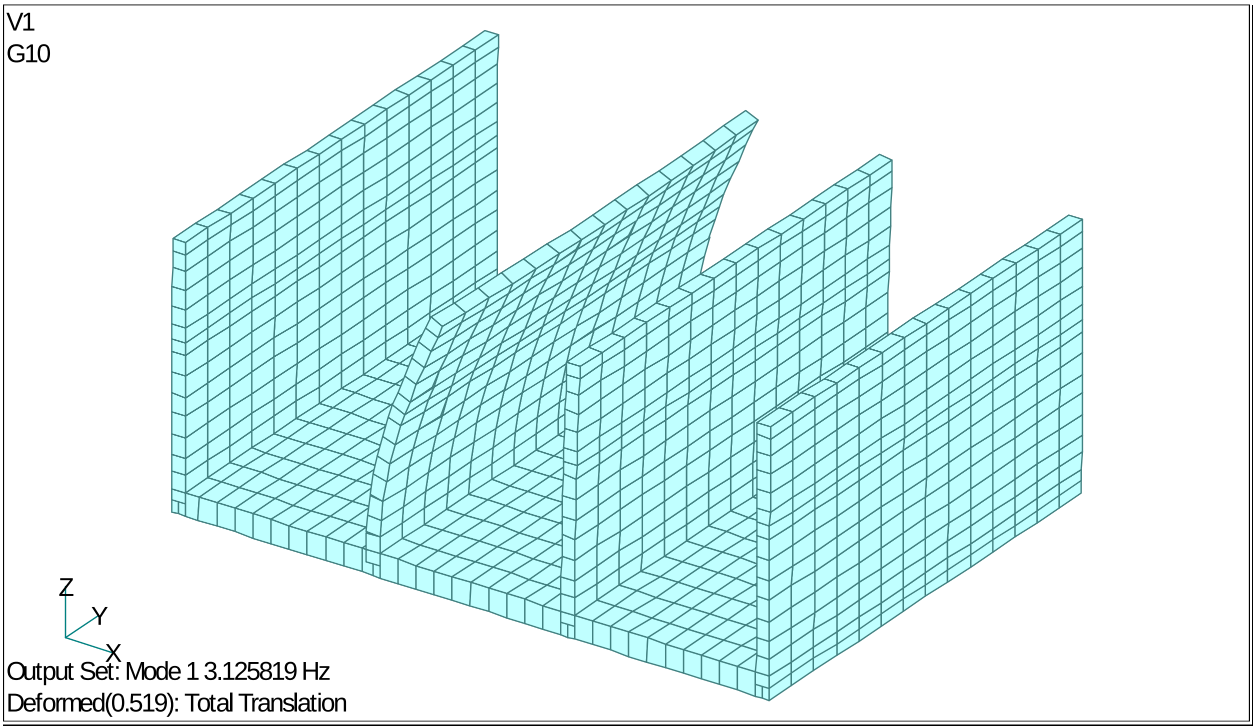


Figure 3.1 First Natural Vibration Mode (First Bending Mode of the Partition Wall, f_1 = 3.126 Hz)

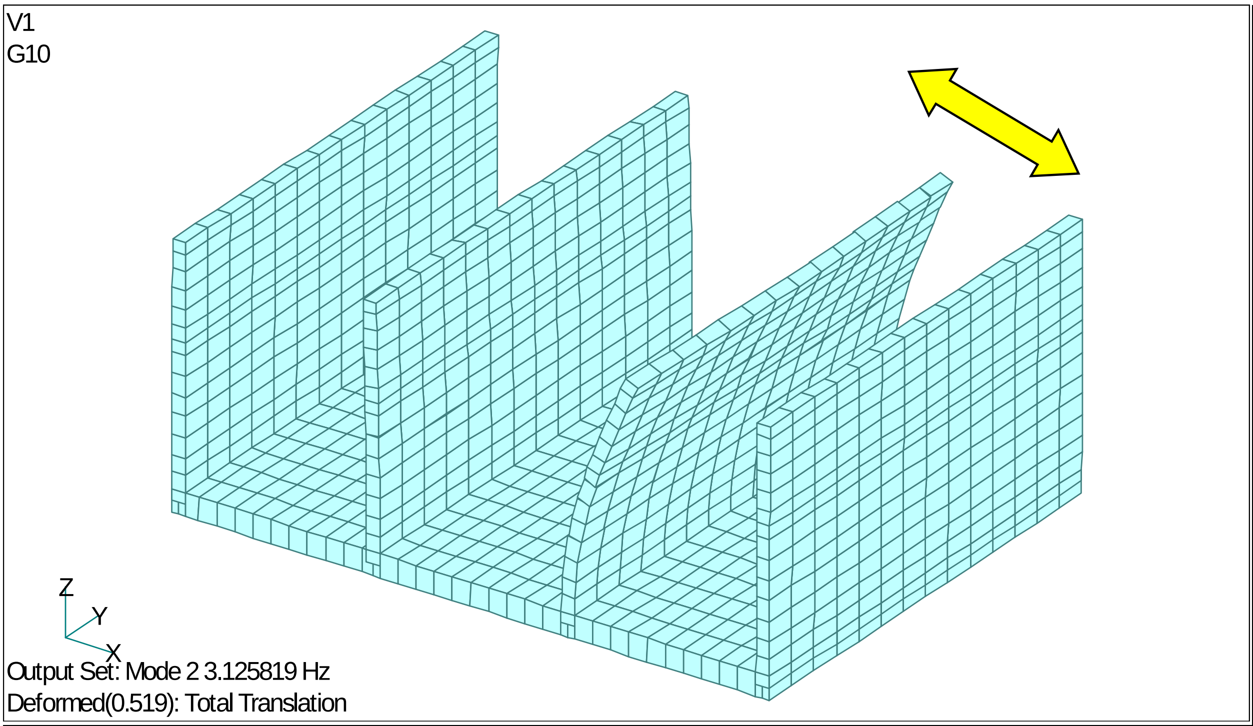


Figure 3.2 Second Natural Vibration Mode (First Bending Mode of the Partition Wall, f_2 = 3.126 Hz)

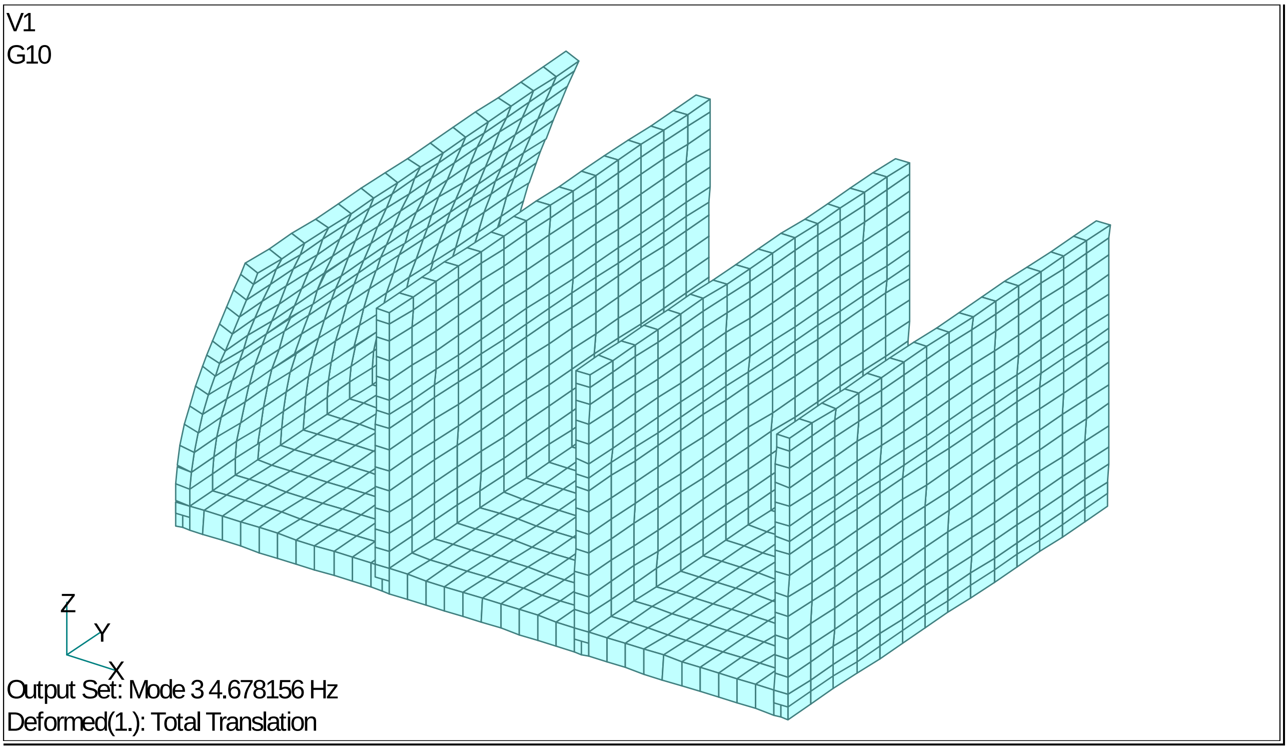


Figure 3.3 Third Natural Vibration Mode (First Bending Mode of the Exterior Wall, f_3 = 4.678 Hz)

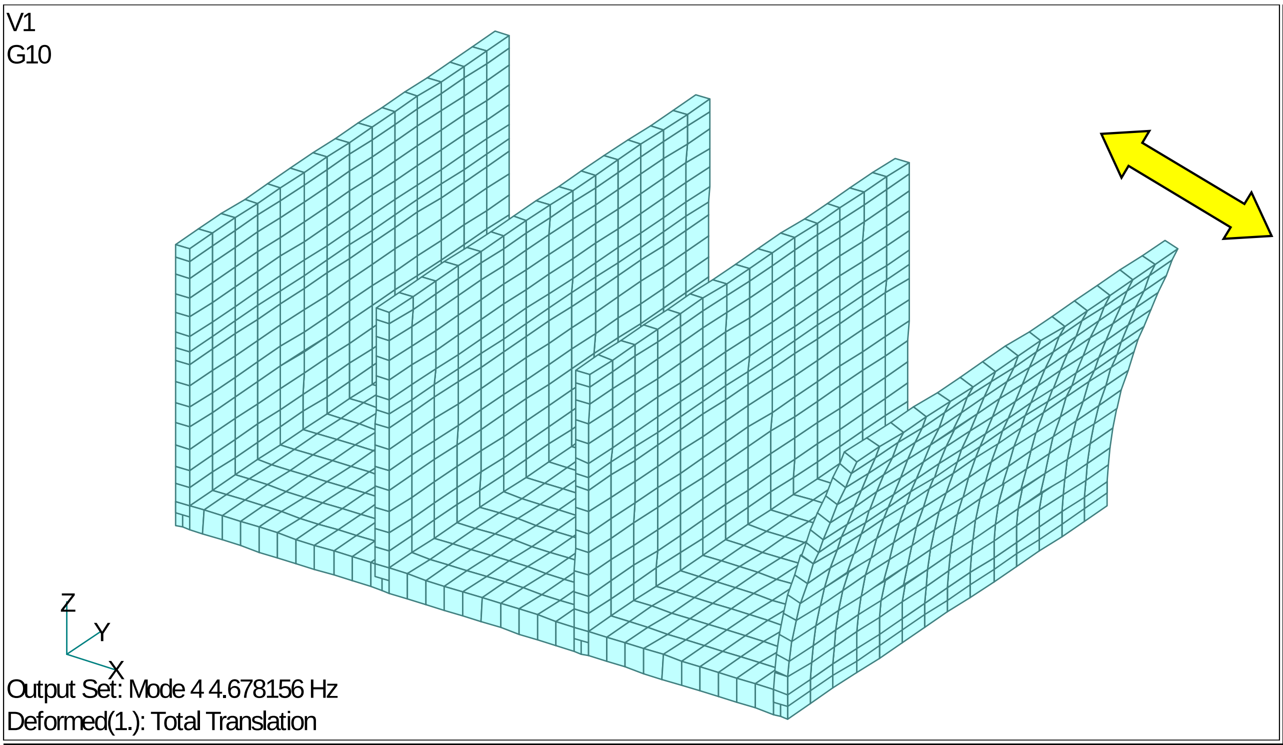


Figure 3.4 Fourth Natural Vibration Mode (First Bending Mode of the Exterior Wall, f_4 = 4.678 Hz)

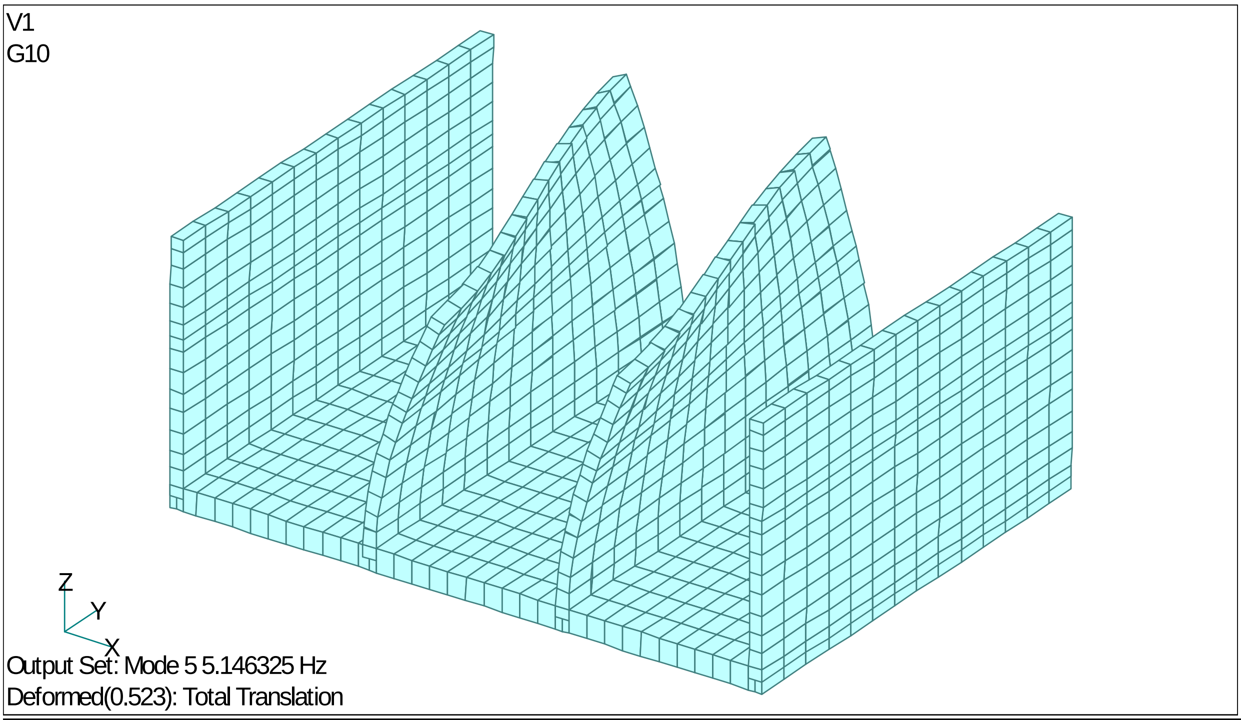


Figure 3.5 Fifth Natural Vibration Mode (First Torsional Mode of the Partition Walls, f_5 = 5.146 Hz; the Two Partition Walls Undergo Torsional Vibration in Phase)

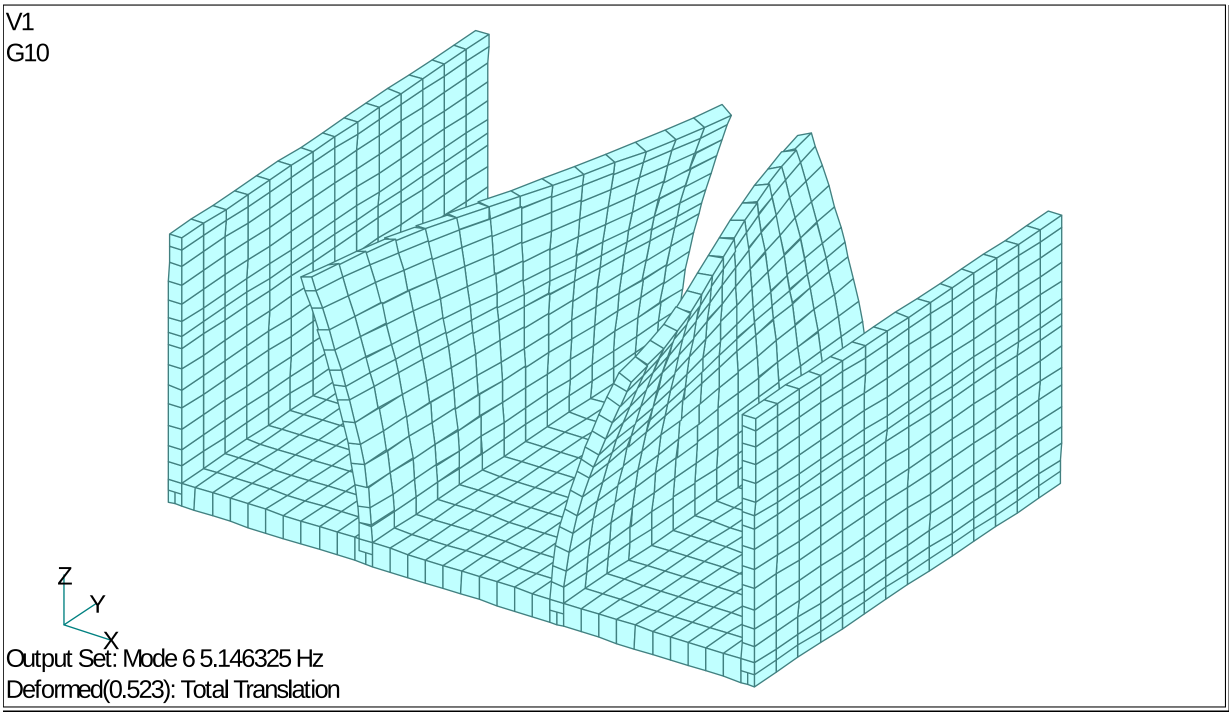


Figure 3.6 Sixth Natural Vibration Mode (Second Torsional Mode of the Partition Walls, f_6 = 5.146 Hz; the Two Partition Walls Undergo Torsional Vibration Out of Phase)

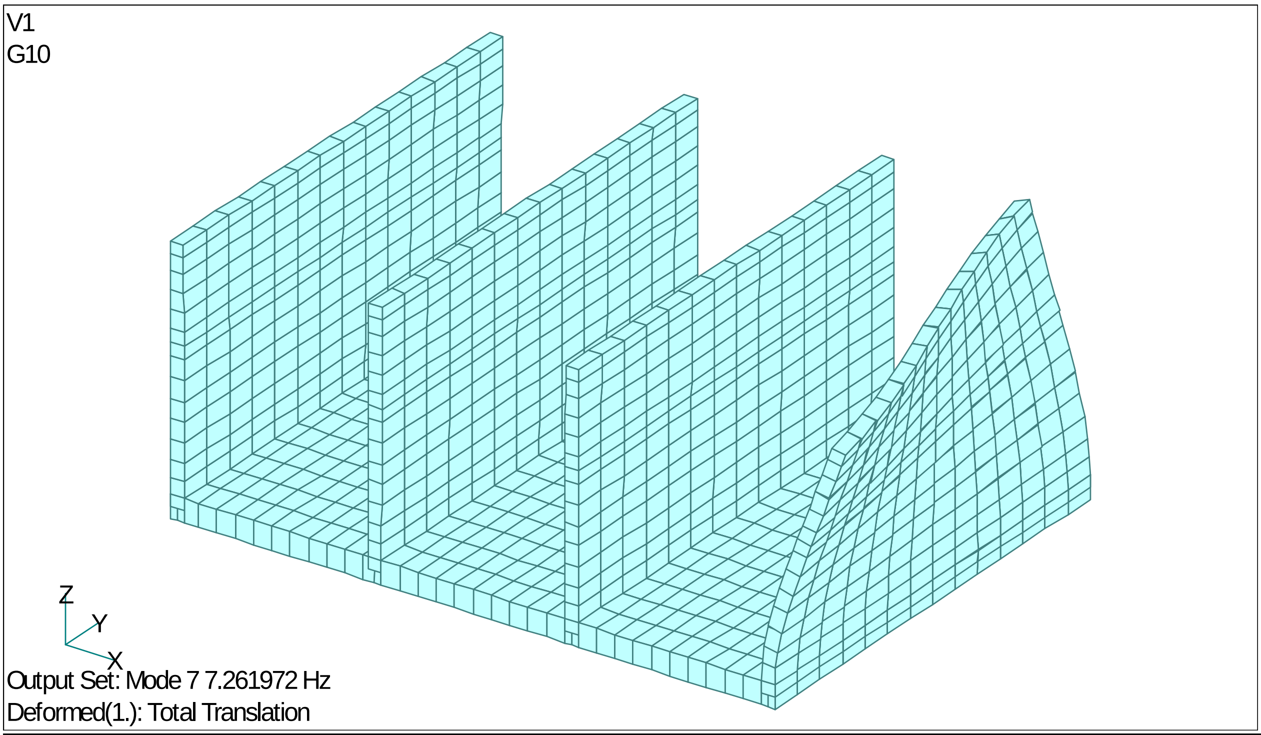


Figure 3.7 Seventh Natural Vibration Mode (First Torsional Mode of the Exterior Walls, f_7 = 7.262 Hz)

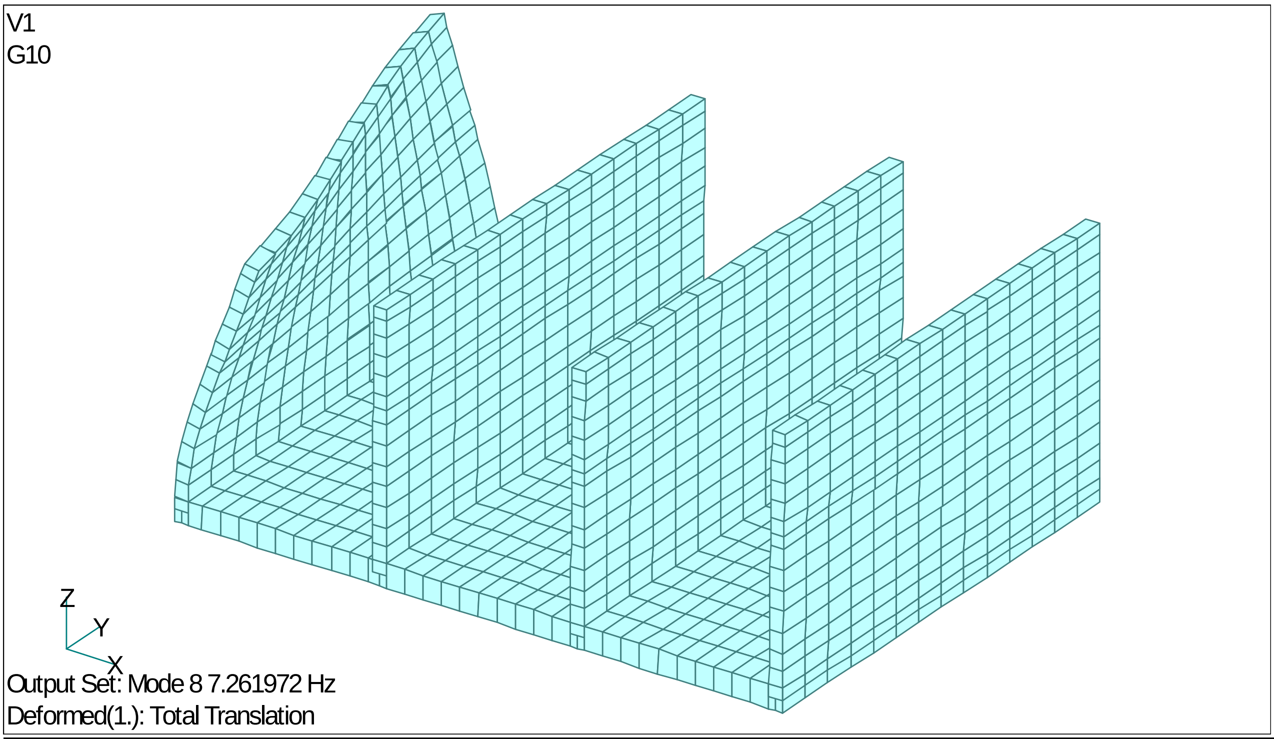


Figure 3.8 Eighth Natural Vibration Mode (First Torsional Mode of the Exterior Walls, f_8 = 7.262 Hz)

Table 3.1 Eigenvalue Analysis Results

Mode	Natural Frequency f (Hz)	Participation Factor X (Weak Axis)	Participation Factor Y (Strong Axis)	Participation Factor Z (Vertical)	Vibration Mode Characteristics
1	3.1258	1.9825	-1.68E-17	-7.48E-18	First bending mode of partition wall
2	3.1258	1.7204	-1.05E-16	2.04E-17	First bending mode of partition wall
3	4.6782	1.4976	-9.84E-17	1.34E-17	First bending mode of exterior wall
4	4.6782	1.4702	5.06E-17	-2.30E-17	First bending mode of exterior wall
5	5.1463	2.24E-02	-3.90E-17	2.46E-17	First torsional mode of partition walls (in-phase)
6	5.1463	1.23E-03	1.76E-16	2.50E-18	Second torsional mode of partition walls (out-of-phase)
7	7.2620	3.87E-04	1.22E-16	-5.65E-17	First torsional mode of exterior wall
8	7.2620	4.12E-04	-3.20E-17	2.60E-17	First torsional mode of exterior wall
9	9.1584	7.16E-02	-2.01E-16	-8.27E-17	Second bending mode of partition wall
10	9.1584	0.13104	-5.25E-16	2.41E-17	Second bending mode of partition wall
11	13.910	6.82E-02	-7.59E-14	2.56E-16	Second bending mode of exterior wall
12	13.910	6.67E-02	2.72E-14	-8.46E-17	Second bending mode of exterior wall

3.2 Seismic Response Analysis Results

A static self-weight analysis followed by a dynamic nonlinear seismic response analysis using the El Centro NS ground motion was performed for the two-story RC Upright Construction Method building.

Figure 3.8 shows the time history of the horizontal relative displacement between the top of the exterior wall and the slab. The maximum horizontal relative displacement between the exterior wall top and the slab is 12.2 mm (at t = 2.93 s).

Figure 3.9 shows the time history of the horizontal acceleration at the top of the exterior wall. The maximum horizontal acceleration at the exterior wall top is 1,098 gal (at t = 3.14 s).

Figure 3.10 shows the time history of the horizontal relative displacement between the top of the interior wall and the slab. The maximum horizontal relative displacement between the interior wall top and the slab is 26.4 mm (at t = 5.02 s).

Figure 3.11 shows the time history of the horizontal acceleration at the top of the interior wall. The maximum horizontal acceleration at the interior wall top is 883 gal (at t = 5.31 s).

Comparing the displacement and acceleration time histories of the exterior wall top shown in Figures 3.8 and 3.9 with those of the interior wall top shown in Figures 3.10 and 3.11, it is considered that the response period of the displacement and acceleration waveforms at the interior wall top is slightly longer, due to the significantly larger roof mass applied at that location.

Figure 3.12 shows the time history of the horizontal relative displacement between the interior wall top and the exterior wall top. The maximum horizontal relative displacement between the interior wall top and the exterior wall top is 28.3 mm (at t = 2.49 s).

Figure 3.13 shows the time history of the horizontal relative displacement between the interior wall at the second-floor level and the exterior wall at the second-floor level. The maximum horizontal relative displacement between the interior wall and the exterior wall at the second-floor level is 10.63 mm (at t = 2.65 s). Note that 100 mm × 100 mm L-shaped steel sections are attached to the walls at the second-floor level of the two-story RC Upright Construction Method building, and the second-floor deck is connected to these L-shaped steel sections. When subjected to the El Centro NS ground motion, the maximum horizontal relative displacement between the interior and exterior walls at the second-floor level is 10.63 mm, which is less than 100 mm. However, since the occurrence of long-period and large-magnitude earthquake motions must also be anticipated in the future, measures to prevent the falling of the second floor (for example, connecting the second-floor deck to the L-shaped steel sections using chains of sufficient strength) are also important from the perspective of occupant safety.

Figure 3.14 shows the deformed shape at the instant of maximum horizontal relative displacement between the exterior wall top and the slab (maximum relative displacement: 12.2 mm, at t = 2.93 s). As shown in the figure, the exterior walls and interior walls deform in phase with each other.

Figure 3.15 shows the deformed shape at the instant of maximum horizontal relative displacement between the interior wall top and the slab (maximum relative displacement: 26.4 mm, at t = 5.02 s). As shown in the figure, at t = 5.02 s, the two interior walls deform in phase with each other, while the exterior walls and interior walls deform out of phase, indicating that the horizontal relative displacement between the interior wall top and the exterior wall top becomes large.

Furthermore, when the El Centro NS ground motion (peak acceleration: 341.7 gal, peak velocity: 33.5 kine, maximum acceleration response spectrum: approximately 700 gal equivalent) was applied to the two-story RC Upright Construction Method building, the analysis results showed that the maximum horizontal relative displacement between the exterior wall top and the slab was approximately 26 mm, which is relatively small, the cracking damage in the walls was extremely minor, and the reinforcing bars did not yield, confirming that the building remains structurally sound.

The reason why the building remains sound even when subjected to the El Centro NS ground motion is considered to be as follows: whereas columns in a typical low-rise RC building are approximately 450 mm square, the two-story RC Upright Construction Method building is equivalent to having 70 columns of 300 mm square cross-section, providing extremely high strength. In addition, all structural elements other than the main RC structure are timber construction, which means the load carried by the main structure is relatively small.

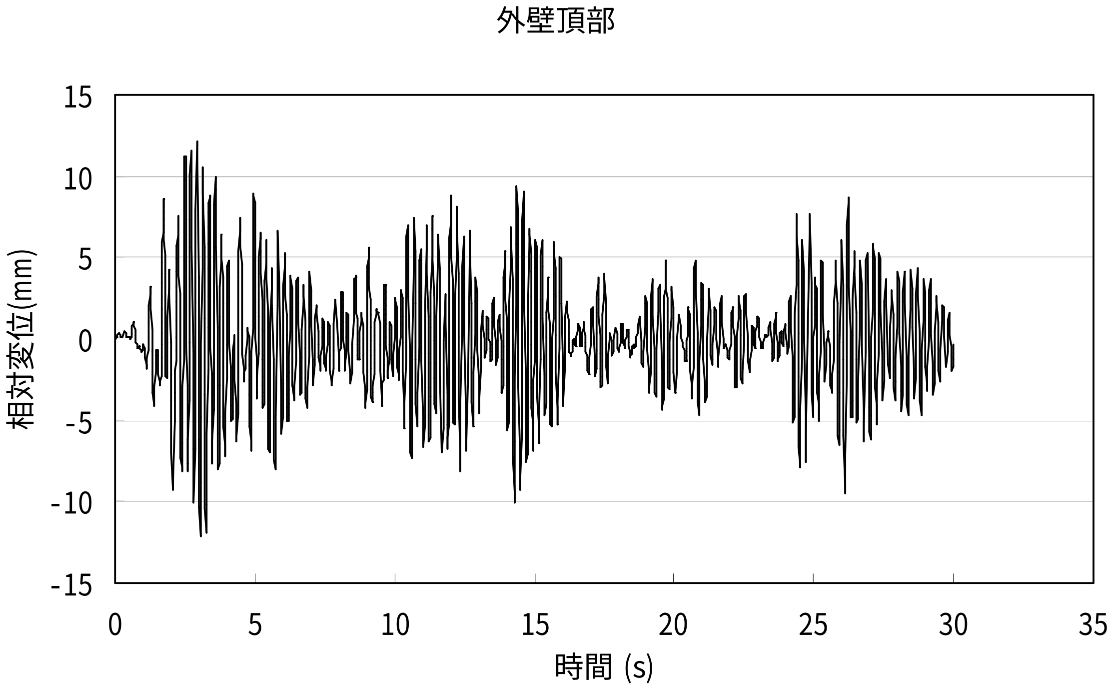


Figure 3.8 Time History of Horizontal Relative Displacement Between the Top of the Exterior Wall and the Slab (Maximum Relative Displacement = 12.2 mm at t = 2.93 s)

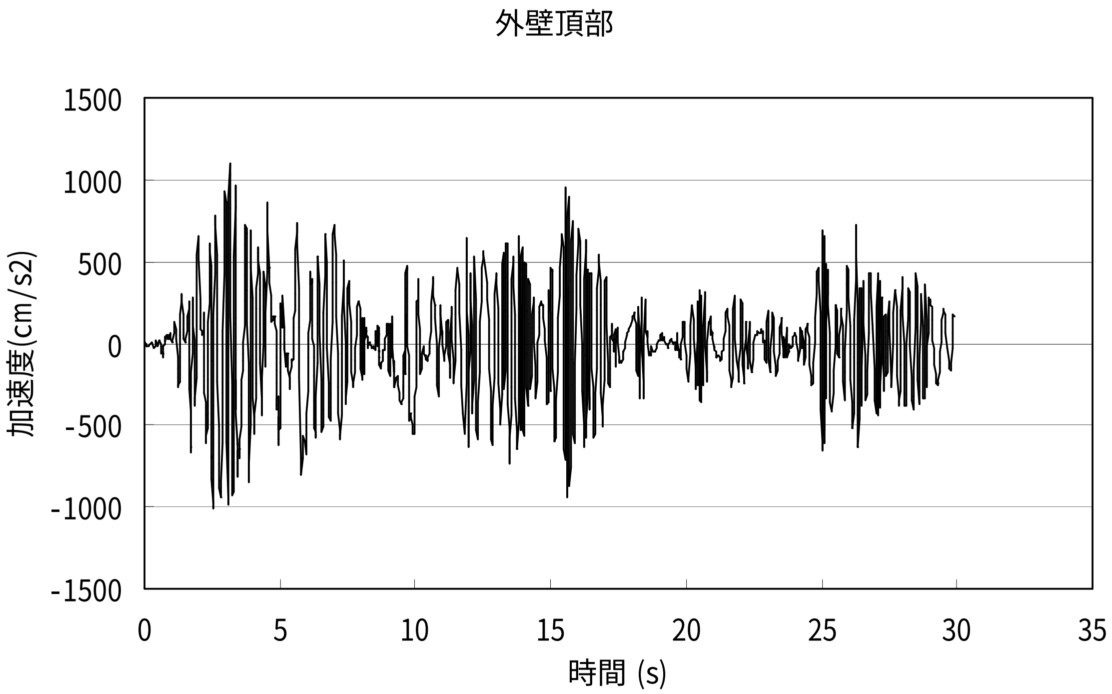


Figure 3.9 Time History of Horizontal Acceleration at the Top of the Exterior Wall (Maximum Acceleration = 1,098 gal at t = 3.14 s)

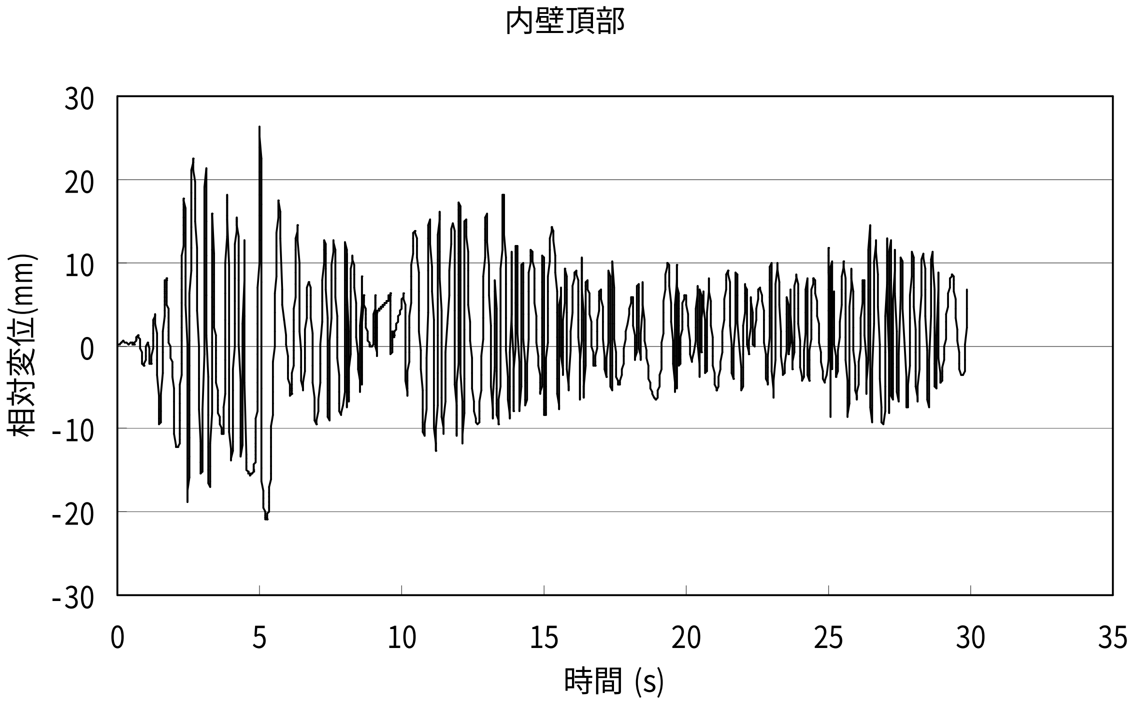


Figure 3.10 Time History of Horizontal Relative Displacement Between the Top of the Partition Wall and the Slab (Maximum Relative Displacement = 26.4 mm at t = 5.02 s)

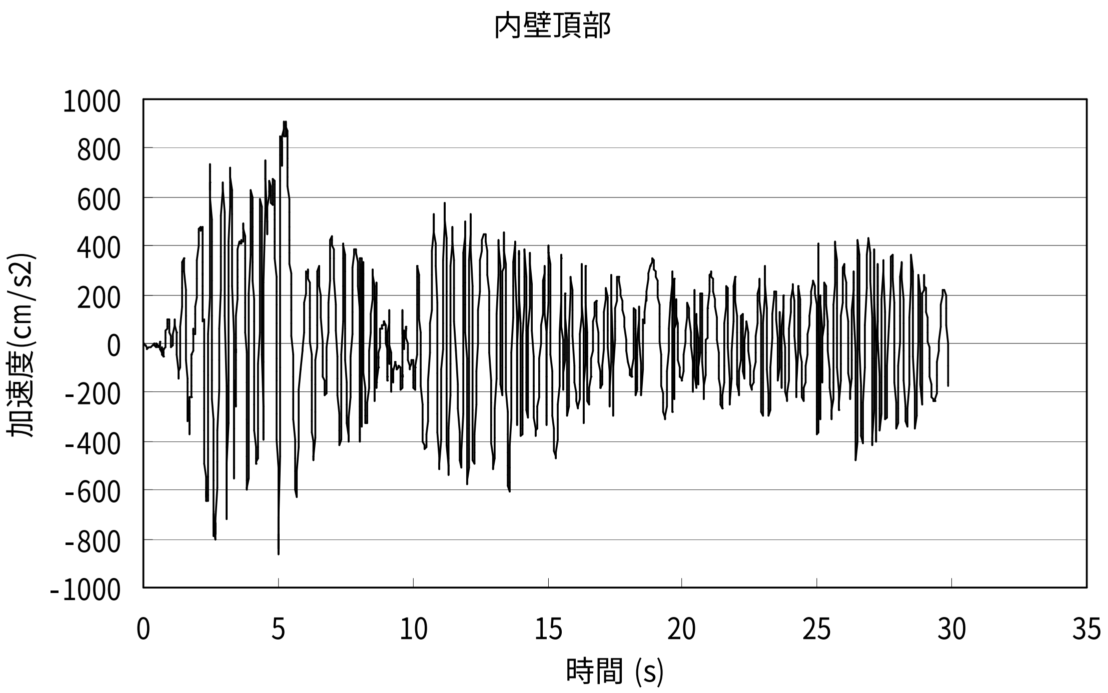


Figure 3.11 Time History of Horizontal Acceleration at the Top of the Partition Wall (Maximum Acceleration = 883 gal at t = 5.31 s)

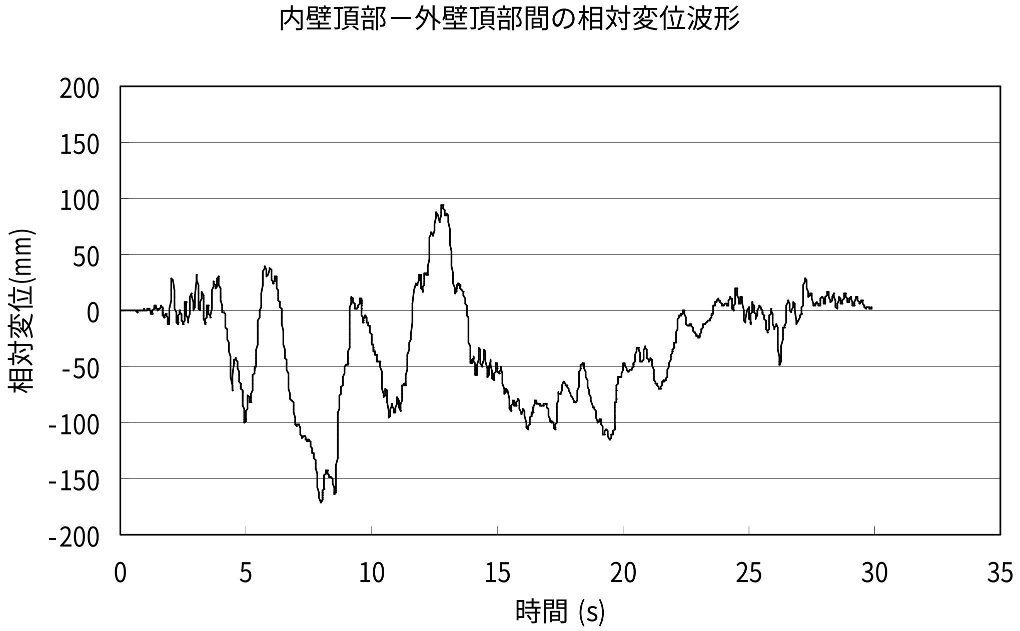


Figure 3.12 Time History of Horizontal Relative Displacement Between the Top of the Partition Wall and the Top of the Exterior Wall (Maximum Relative Displacement = 164.6 mm at t = 8.04 s)

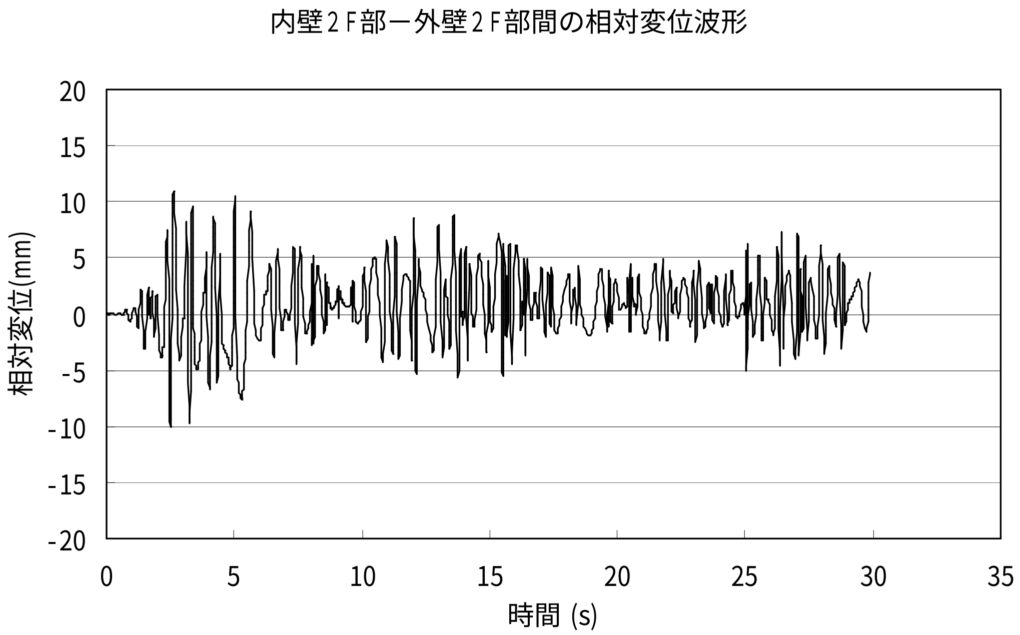


Figure 3.13 Time History of Horizontal Relative Displacement Between the Second-Floor Portions of the Partition Wall and the Exterior Wall (Maximum Relative Displacement = 10.63 mm at t = 2.65 s)

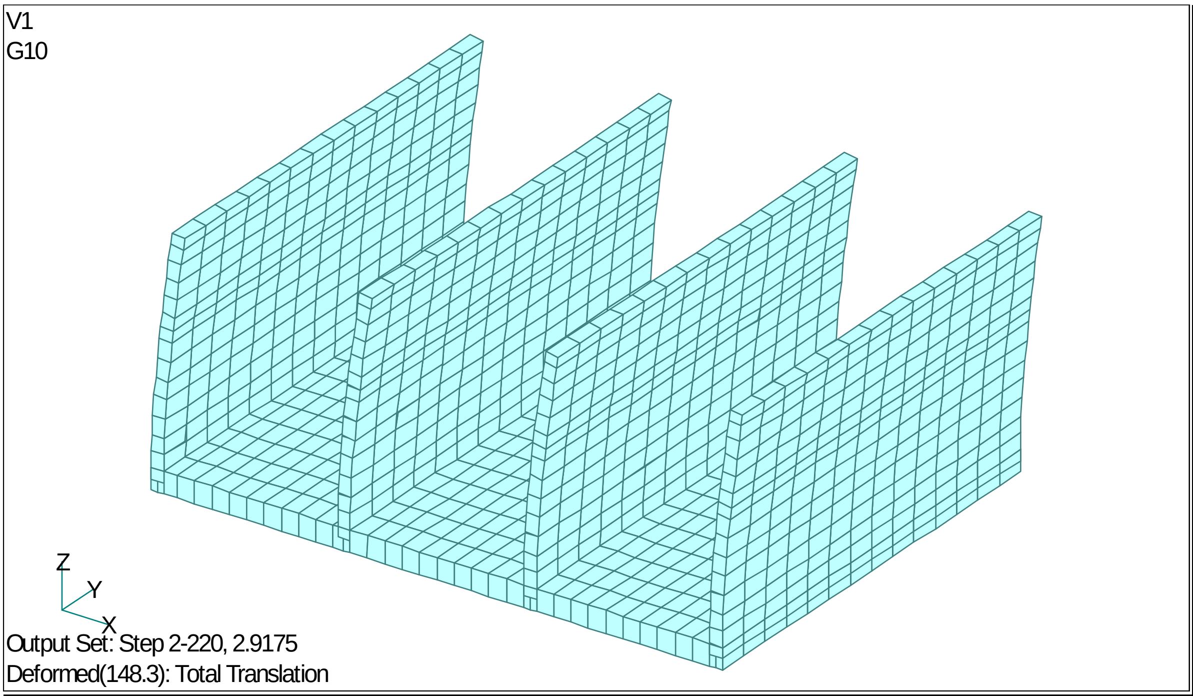


Figure 3.14 Deformed Shape at the Time of Maximum Horizontal Relative Displacement Between the Top of the Exterior Wall and the Slab (Maximum Relative Displacement = 12.2 mm at t = 2.93 s)

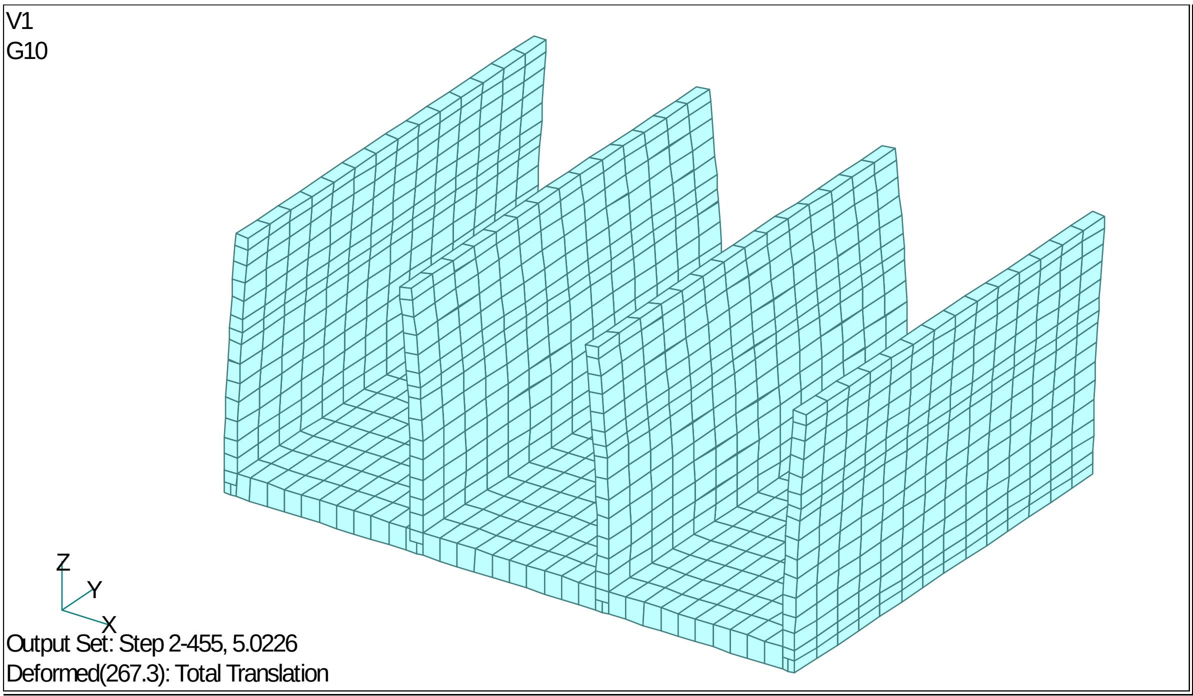


Figure 3.15 Deformed Shape at the Time of Maximum Horizontal Relative Displacement Between the Top of the Partition Wall and the Slab (Maximum Relative Displacement = 26.4 mm at t = 5.02 s)

4. Summary

In this study, the two-story RC Upright Construction Method building was modeled using three-dimensional thick-shell finite elements incorporating reinforcing bars, and a dynamic nonlinear seismic response analysis was performed to evaluate its structural integrity under earthquake loading. The principal findings obtained from this analysis, together with topics requiring further investigation, are summarized below.

(1) According to the eigenvalue analysis results of the two-story RC Upright Construction Method building, the first natural frequency of the building was f_1 = 3.13 Hz (T_1 = 0.32 s). In addition, according to a separate field measurement survey conducted on the subject building, in which a vibration exciter was installed on the second floor to measure the natural frequency, the first natural frequency of the building was f_1 = 6.2 Hz (T_1 = 0.16 s), indicating that the building has a relatively stiff structural system. The first natural frequency obtained from the present eigenvalue analysis, f_1 = 3.13 Hz (T_1 = 0.32 s), is lower than the measured value. This is considered to be due to the fact that, in this analysis, the stiffness of the roof, wooden walls, and floors was neglected, resulting in a lower predicted first natural frequency.

(2) When the El Centro NS ground motion (peak acceleration: 341.7 gal, peak velocity: 33.5 kine, maximum acceleration response spectrum: approximately 700 gal equivalent) was applied to the two-story RC Upright Construction Method building, the analysis results showed that the maximum horizontal relative displacement between the exterior wall top and the slab was approximately 13 mm, which is relatively small, the cracking damage in the walls was extremely minor, and the reinforcing bars did not yield, confirming that the building remains structurally sound. The reason why the building remains sound even when subjected to the El Centro NS ground motion is considered to be as follows: whereas columns in a typical low-rise RC building are approximately 450 mm square, the two-story RC Upright Construction Method building is equivalent to having 70 columns of 300 mm square cross-section, providing extremely high structural strength. In addition, all structural elements other than the main RC structure are timber construction, which means the load carried by the main structure is relatively small.

(3) As topics for future study, the weight reduction of RC Upright Construction Method buildings should be investigated, including the possibility of increasing the wall thickness at the lower portion of the wall while introducing a tapered section to reduce the wall thickness toward the upper portion. It is also considered important to perform large-deformation elasto-plastic analyses of RC Upright Construction Method buildings under progressively increasing horizontal seismic accelerations in order to determine the ultimate acceleration capacity. Based on such results, additional dynamic nonlinear seismic response analyses using stronger ground motions and earthquake records with different frequency characteristics should also be conducted to further evaluate the structural integrity under earthquake loading. From a construction perspective, measures to prevent the falling of second-floor structural components during earthquakes are also important from the standpoint of occupant safety. Furthermore, the installation of base-isolation and seismic energy dissipation (vibration control) devices effective against the large-scale earthquakes anticipated in the future, as well as the development of base-isolation and vibration control devices specifically suited to RC Upright Construction Method buildings, are also considered important.

Appendices

Appendix 1 Additional Masses Applied to Walls and Slabs

Appendix 2 Representative Recorded Earthquake Motions (Acceleration Data)

Appendix 3 Design Response Spectra for Horizontal Seismic Motions

Reference: Building Research Institute, Ministry of Construction, and Building Center of Japan, Technical Guidelines (Draft) for the Generation of Design Input Ground Motions — Commentary Edition, March 1992.

In the design velocity response spectra (Level 1 and Level 2) shown on the following pages, a velocity of 100 kine corresponds to an acceleration of 1,000 gal in terms of the acceleration response spectrum. The thick solid line in the figures represents the velocity response spectrum of the design prototype wave B(T). The acceleration time history, velocity time history, and displacement time history of the design prototype wave B(T) (Level 1 and Level 2) are shown on the subsequent pages. The maximum acceleration of the prototype wave B(T) in the acceleration time history is 207.3 gal (Level 1) and 355.7 gal (Level 2); however, in the respective design velocity response spectra, the maximum values are 50 kine (Level 1, equivalent to 500 gal) and 100 kine (Level 2, equivalent to 1,000 gal). The design velocity response spectra (Level 1 and Level 2) also include the velocity response spectra of the El Centro NS wave scaled to amplitudes of 20 kine (Level 1) and 50 kine (Level 2), shown as dashed lines with yellow markers. The El Centro NS wave used in the present seismic response analysis (341.7 gal, 33.5 kine) is considered to correspond to a maximum acceleration spectrum of approximately 700 gal.

For reference, the acceleration response spectra (h = 5%) and velocity response spectra (h = 5%) of the 1995 Hyogo-ken Nanbu Earthquake and the 2004 Niigata Chuetsu Earthquake, based on data from the Disaster Prevention Research Institute, Kyoto University (Iwata), are also shown. As shown in the figures, the Niigata Chuetsu Earthquake exhibited an extremely large maximum acceleration spectrum of 6,000 gal (h = 5%) and a maximum velocity spectrum of 600 kine (h = 5%), significantly exceeding those of the Hyogo-ken Nanbu Earthquake.

Appendix 4 ABAQUS Concrete Model

References

1) Architectural Institute of Japan, Technical Guidelines for High-Rise Buildings – Revised and Expanded Edition, August 1976.

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3) Architectural Institute of Japan, Report on the Hanshin-Awaji Earthquake Disaster: Building Series Vol. 2 – Prestressed Concrete Buildings, Steel Reinforced Concrete Buildings, and Wall-Structure Buildings, August 1998.

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